

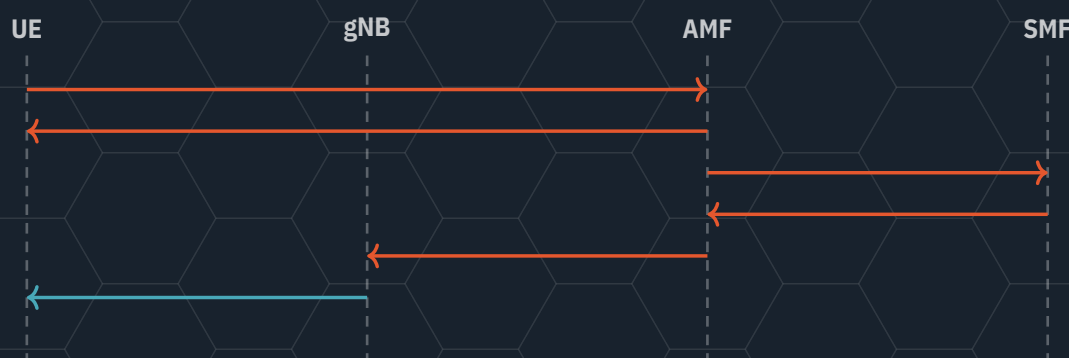
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 20

5G Procedures

رویه‌های 5G



Lecture Notes

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1 Registration (Initial Registration)

Purpose

The Initial Registration procedure allows a UE to register with the 5G Core Network (5GC) for the first time or after being deregistered. It establishes the UE's presence in the network, authenticates the subscriber, allocates a 5G-GUTI (Globally Unique Temporary Identifier), and optionally establishes a PDU session for data connectivity.

Preconditions

- UE is powered on and has a valid USIM with 5G subscription credentials
- UE has selected a PLMN and camped on a cell (gNB)
- UE is in RRC_IDLE or RRC_INACTIVE state
- UE has no valid 5G-GUTI (first-time registration) or previous GUTI has expired
- RRC connection has been established (Random Access procedure completed)

NFs Involved

Network Function	Role
UE	Initiates registration, provides identity (SUCI/5G-GUTI)
gNB	RAN node, forwards NAS messages between UE and AMF
AMF	Anchor for registration, selects AUSF, manages context
AUSF	Authentication server, validates credentials
UDM	Stores subscription data, generates auth vectors
ARPF	Authentication credential Repository (within UDM)
PCF	Provides AM policies
SMF	Session management (if PDU session requested)
UPF	User plane (if PDU session requested)
NRF	NF discovery and selection
NSSF	Network Slice selection

Messages

Step	Interface	Message
1	Uu (UE→gNB)	RRCSetupRequest
2	Uu (gNB→UE)	RRCSetup
3	Uu (UE→gNB)	RRCSetupComplete + NAS Registration Request
4	N2 (gNB→AMF)	NGAP Initial UE Message (NAS: Registration Request)
5	N12 (AMF→AUSF)	Nausf_UEAuthentication_Authenticate Request
6	N13 (AUSF→UDM)	Nudm_UEAuthentication_Get Request
7	N13 (UDM→AUSF)	Nudm_UEAuthentication_Get Response (AV)
8	N12 (AUSF→AMF)	Nausf_UEAuthentication_Authenticate Response
9	N2/NAS (AMF→UE)	Authentication Request (RAND, AUTN)
10	N2/NAS (UE→AMF)	Authentication Response (RES*)
11	N12 (AMF→AUSF)	Nausf_UEAuthentication_Authenticate (RES*)
12	N12 (AUSF→AMF)	Nausf_UEAuthentication_Authenticate (Success + K_{SEAF})
13	N2/NAS (AMF→UE)	Security Mode Command
14	N2/NAS (UE→AMF)	Security Mode Complete

Step	Interface	Message
15	N8 (AMF→UDM)	Nudm_UECM_Registration
16	N8 (AMF→UDM)	Nudm_SDM_Get (Subscription Data)
17	N15 (AMF→PCF)	Npcf_AMPolicyControl_Create
18	N2/NAS (AMF→UE)	Registration Accept (5G-GUTI, TAI List, Allowed NSSAI)
19	N2/NAS (UE→AMF)	Registration Complete
20	N2 (AMF→gNB)	NGAP Initial Context Setup Request

Step-by-Step Flow

- RRC Connection Establishment:** UE performs Random Access and sends RRCSetupRequest to gNB
- RRC Setup:** gNB responds with RRCSetup message
- Registration Request:** UE sends RRCSetupComplete containing NAS Registration Request (includes SUCI or 5G-GUTI, Registration Type, UE Security Capabilities, Requested NSSAI)
- AMF Selection:** gNB selects appropriate AMF (via NSSF if needed) and forwards via NGAP Initial UE Message
- Identity Request (conditional):** If AMF cannot resolve identity, sends Identity Request to UE
- SUCI Resolution:** AMF initiates authentication; sends SUCI to AUSF
- Auth Vector Request:** AUSF requests authentication vector from UDM/ARPF
- AV Generation:** UDM/ARPF generates 5G HE AV (RAND, AUTN, XRES*, K_{AUSF}), resolves SUCI→SUPI
- Auth Vector Response:** AUSF receives AV, computes HXRES, stores XRES
- Authentication Request:** AMF sends NAS Authentication Request to UE (RAND, AUTN)
- UE Verification:** UE verifies AUTN, computes RES*, CK, IK, derives K_{AUSF}
- Authentication Response:** UE sends Authentication Response (RES*) to AMF
- RES* Verification:** AMF sends RES* to AUSF; AUSF verifies HRES* matches HXRES*
- Authentication Confirmation:** AUSF confirms to AMF, provides K_{SEAF} ; AMF derives K_{AMF}
- NAS Security:** AMF sends Security Mode Command (selected algorithms); UE responds with Security Mode Complete
- UDM Registration:** AMF registers with UDM (Nudm_UECM_Registration)
- Subscription Retrieval:** AMF retrieves subscription data from UDM (Nudm_SDM_Get)
- Policy Association:** AMF establishes AM Policy association with PCF
- Registration Accept:** AMF sends Registration Accept (5G-GUTI, TAI List, Allowed NSSAI, Registration Area)
- Registration Complete:** UE acknowledges with Registration Complete; stores new 5G-GUTI

Detailed Sequence Diagram

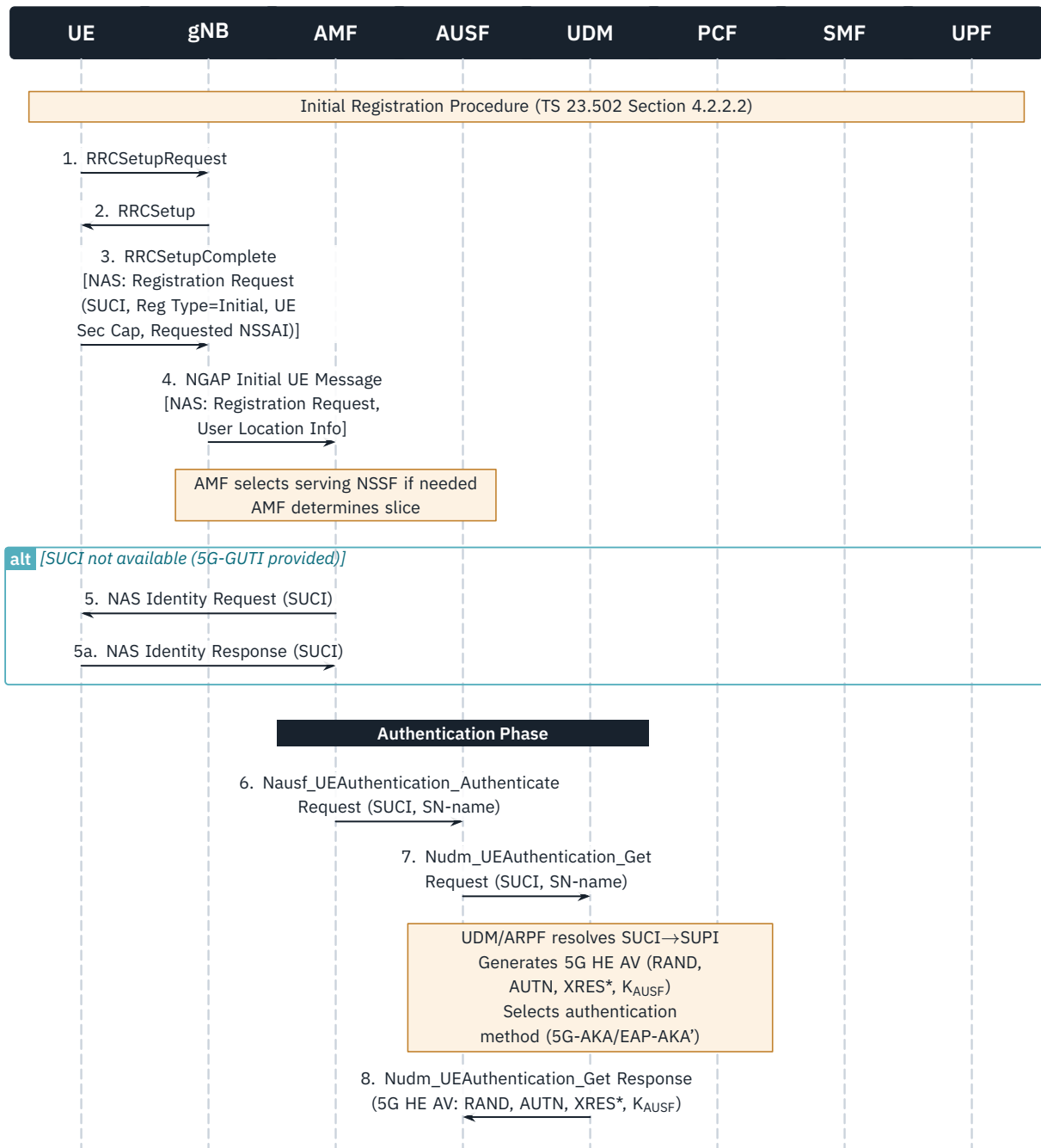


Figure 20.1 Initial registration (TS 23.502 §4.2.2.2) (part 1 of 3)

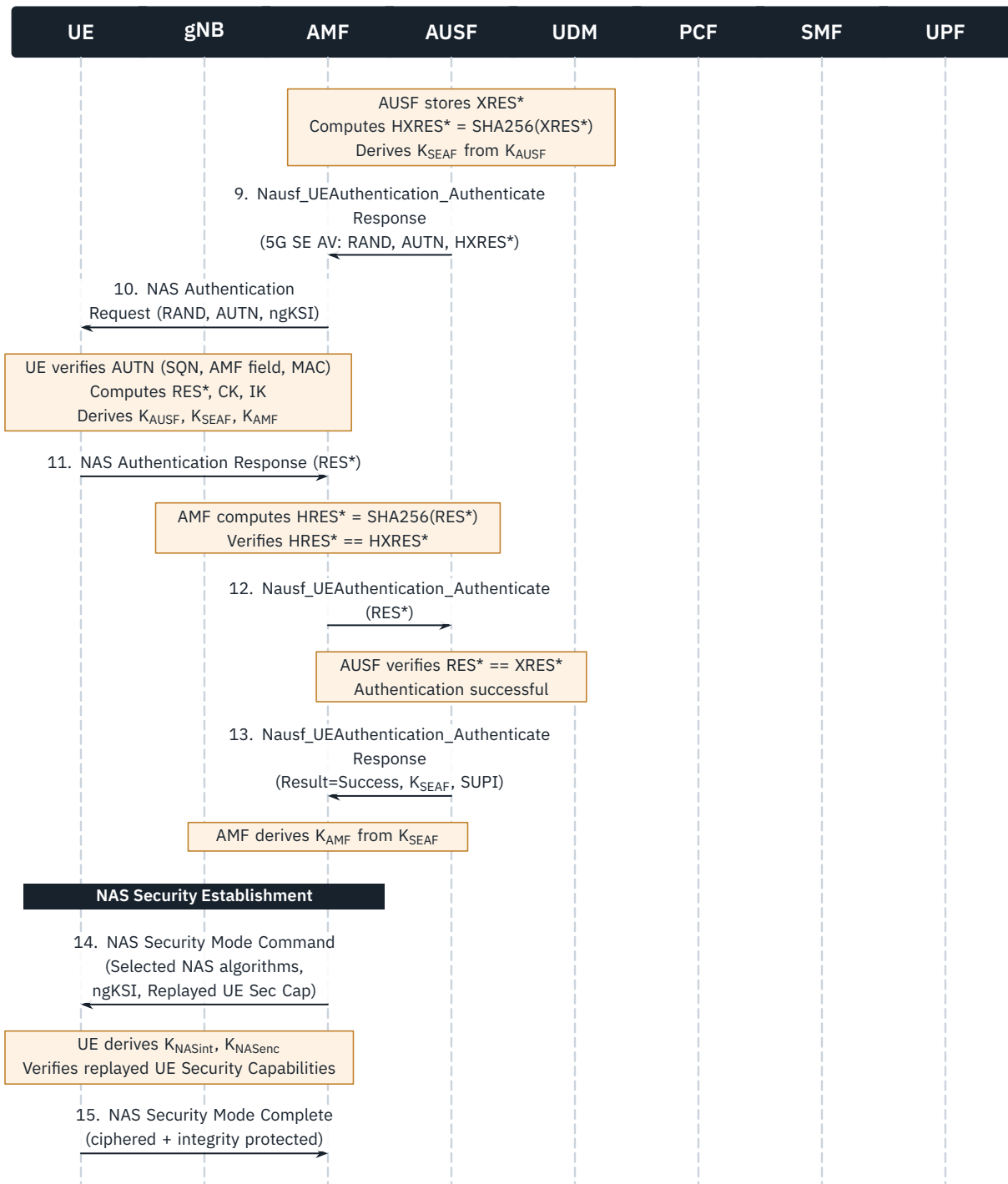


Figure 20.2 Initial registration (TS 23.502 §4.2.2.2) (part 2 of 3)

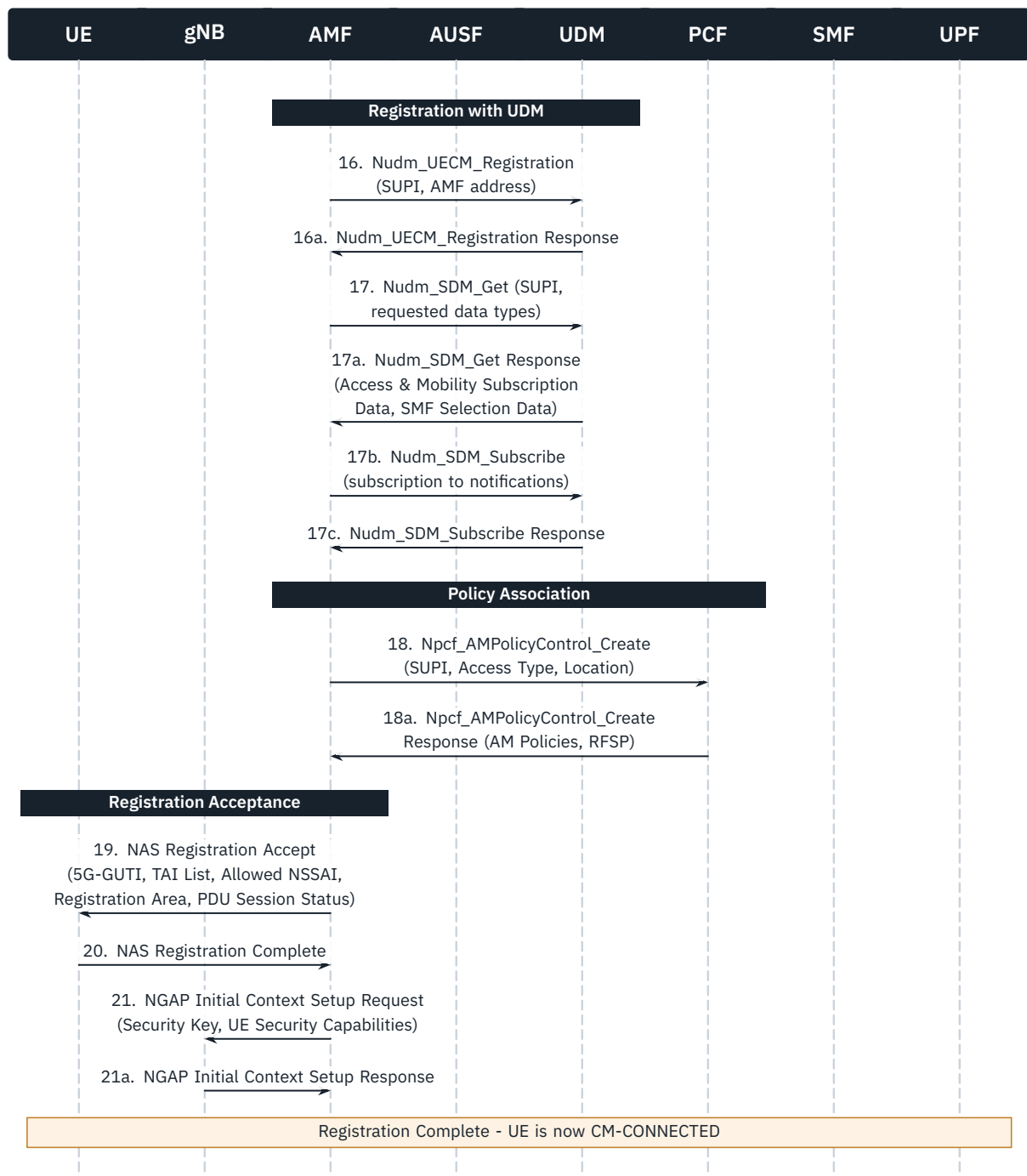


Figure 20.3 Initial registration (TS 23.502 §4.2.2.2) (part 3 of 3)

Failure Cases

Failure	Cause	Network Response
Authentication Failure	Invalid credentials, SQN out of sync	Authentication Reject or Sync Failure
Registration Reject	No subscription, roaming not allowed	Registration Reject (Cause: #11, #15)
PLMN Not Allowed	UE not subscribed for visited PLMN	Registration Reject (Cause: #11)
Congestion	Network overload	Registration Reject (Cause: #22) + back-off timer
Integrity Check Failure	NAS MAC failure	Reject silently or re-authenticate
UDM Unreachable	SBI communication failure	Registration Reject or AMF retry
Slice Not Available	Requested S-NSSAI not supported	Registration Accept with reduced Allowed NSSAI

2 Authentication (5G-AKA)

Purpose

The 5G-AKA (Authentication and Key Agreement) procedure provides mutual authentication between the UE and network, and establishes shared cryptographic keys. It ensures the network verifies the subscriber's identity and the UE verifies it is communicating with a legitimate network. The procedure generates the anchor key (K_{AUSF}) from which all subsequent security keys are derived.

Preconditions

- UE has initiated Registration or Re-authentication is triggered
- UE has valid USIM with long-term key K and OP/OPc
- AUSF is reachable from the serving AMF
- UDM/ARPF has valid subscription credentials for the subscriber
- SUCI has been provided (either from UE or resolved from 5G-GUTI)

■ NFs Involved

Network Function	Role
UE (USIM)	Holds permanent key K ; verifies AUTN; computes RES^*
AMF (SEAF)	Security Anchor; relays auth messages; derives K_{AMF}
AUSF	Verifies RES^* against $XRES^*$; derives K_{SEAF}
UDM/ARPF	Generates authentication vectors; holds permanent key K
SIDF (in UDM)	Decrypts SUCI to obtain SUPI

■ Messages

Step	Interface	Message	Key Parameters
1	N12	Nausf_UEAuthentication_Authenticate Request	SUCI/SUPI, Serving Network Name
2	N13	Nudm_UEAuthentication_Get Request	SUCI/SUPI, Serving Network Name
3	N13	Nudm_UEAuthentication_Get Response	5G HE AV (RAND, AUTN, $XRES^*$, K_{AUSF})
4	N12	Nausf_UEAuthentication_Authenticate Response	5G SE AV (RAND, AUTN, $HXRES^*$)
5	NAS	Authentication Request	RAND, AUTN, ngKSI
6	NAS	Authentication Response	RES^*
7	N12	Nausf_UEAuthentication_Authenticate	RES^*
8	N12	Nausf_UEAuthentication_Authenticate Response	Result, K_{SEAF} , SUPI

■ Step-by-Step Flow

1. **Trigger:** AMF determines authentication is needed (initial registration, periodic re-auth, or inter-AMF handover)
2. **AUSF Selection:** AMF selects AUSF based on SUPI home network (via NRF)
3. **Auth Initiation:** AMF sends SUCI and Serving Network Name (SN-name) to AUSF
4. **UDM Request:** AUSF forwards request to UDM with SUCI and SN-name
5. **SUCI Decryption:** SIDF (within UDM) decrypts SUCI using home network private key → obtains SUPI
6. **AV Generation:** ARPF generates 5G HE AV using:
 - RAND (random challenge)
 - $AUTN = SQN \oplus AK \parallel AMF \parallel MAC$ (network authentication token)
 - $XRES^* = KDF(CK, IK, SN-name, RAND, RES)$ (expected response)
 - $K_{AUSF} = KDF(CK, IK, SN-name, SQN \oplus AK)$ (anchor key)

7. **AV to AUSF:** UDM returns 5G HE AV to AUSF
8. **AUSF Processing:** AUSF stores XRES, computes $HXRES = \text{SHA-256}(XRES[128..255] || XRES)$, derives K_{SEAF} from K_{AUSF}
9. **SE AV to AMF:** AUSF sends 5G SE AV (RAND, AUTN, HXRES*) to AMF
10. **Challenge to UE:** AMF sends Authentication Request (RAND, AUTN, ngKSI) to UE
11. **UE Verification:** USIM verifies AUTN:
 - Extracts AK from RAND using f5, recovers SQN
 - Computes XMAC and verifies $XMAC == MAC$ (network authentication)
 - Verifies SQN is in acceptable range (replay protection)
12. **UE Key Derivation:** UE computes:
 - RES, CK, IK from RAND using f2, f3, f4
 - $RES^* = KDF(CK, IK, SN\text{-}name, RAND, RES)$
 - $K_{AUSF} = KDF(CK, IK, SN\text{-}name, SQN \oplus AK)$
 - $K_{SEAF} = KDF(K_{AUSF}, SN\text{-}name)$
 - $K_{AMF} = KDF(K_{SEAF}, SUPI, ABBA)$
13. **Response:** UE sends Authentication Response with RES* to AMF
14. **AMF Verification:** AMF computes HRES* and verifies $HRES^* == HXRES^*$
15. **AUSF Verification:** AMF forwards RES* to AUSF; AUSF verifies $RES^* == XRES^*$
16. **Success:** AUSF returns success, K_{SEAF} , and confirmed SUPI to AMF
17. **K_{AMF} Derivation:** AMF derives $K_{AMF} = KDF(K_{SEAF}, SUPI, ABBA)$

■ Key Hierarchy Derived

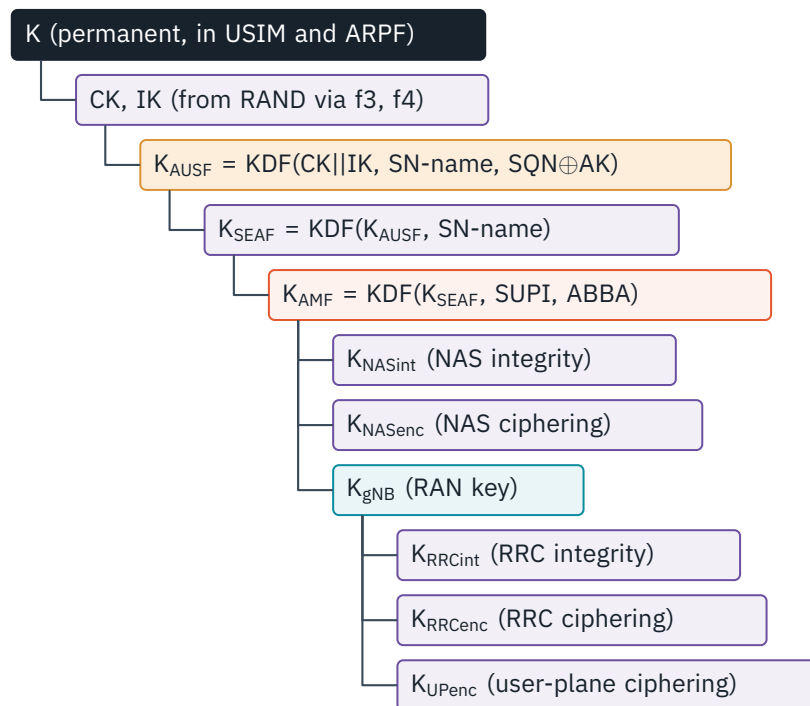


Figure 20.4 Key hierarchy derived during 5G-AKA

Detailed Sequence Diagram

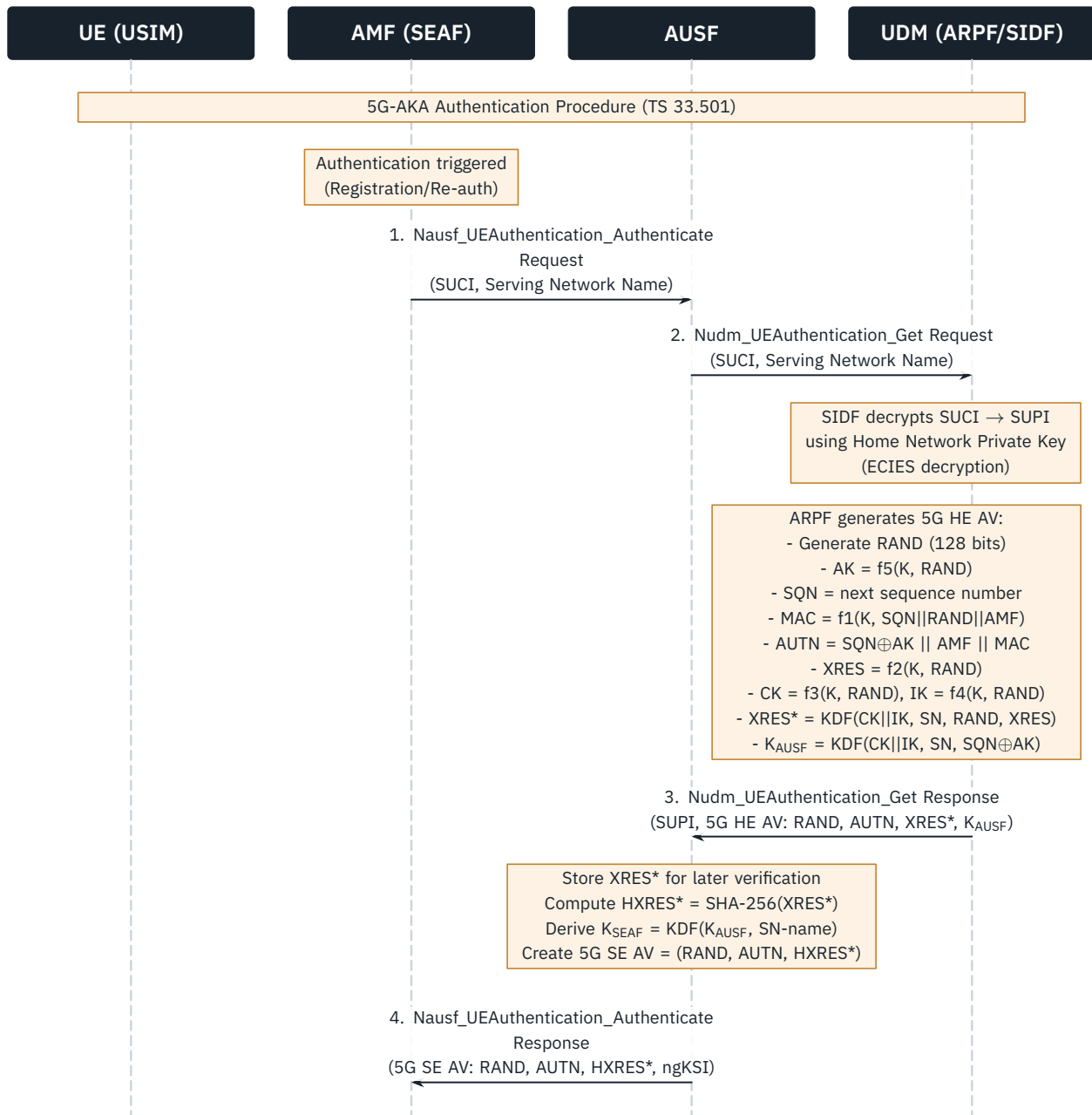


Figure 20.5 5G-AKA authentication (TS 33.501) (part 1 of 2)

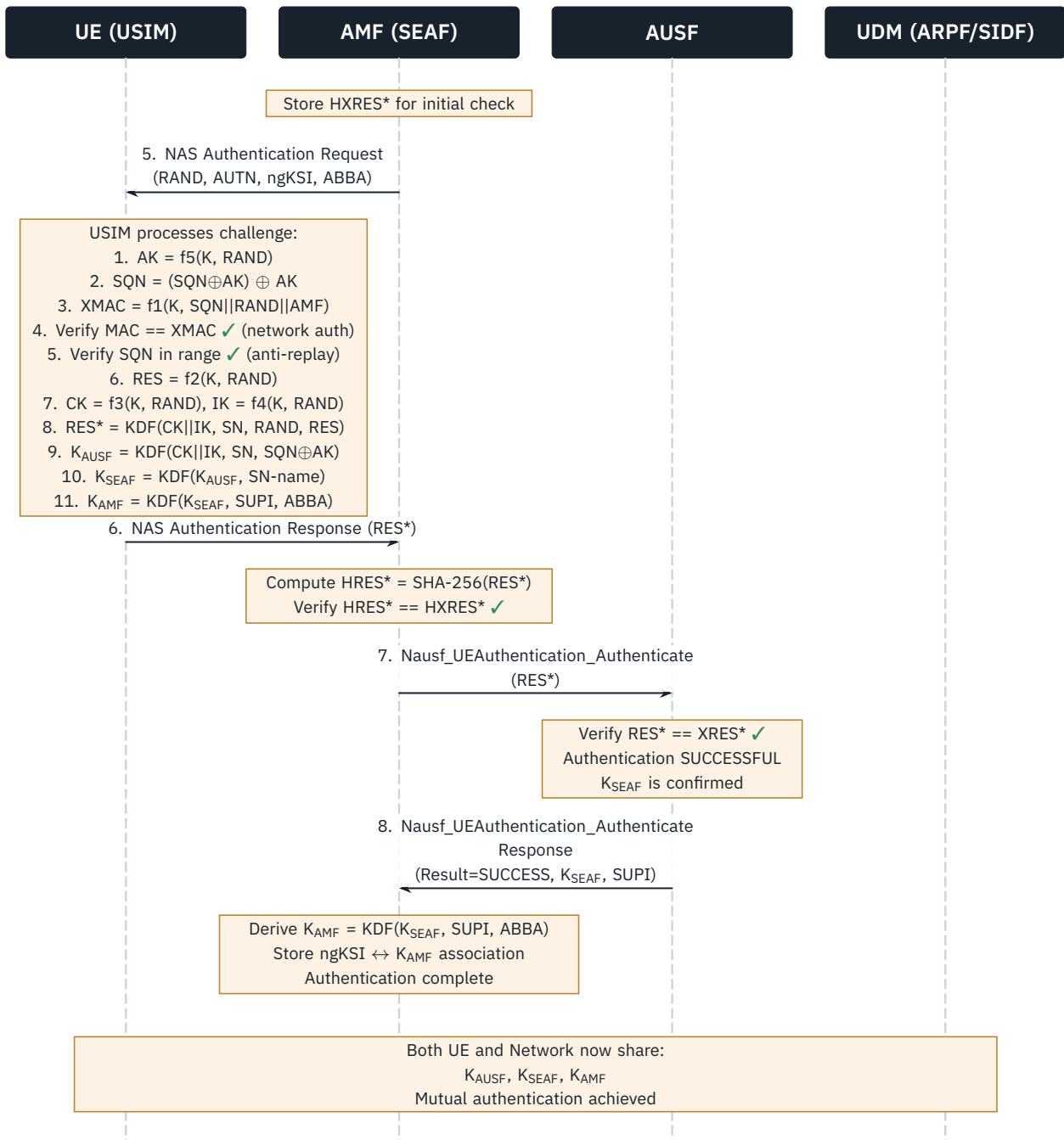


Figure 20.6 5G-AKA authentication (TS 33.501) (part 2 of 2)

Failure Cases

Failure	Cause	UE Action	Network Action
MAC Failure	AUTN verification fails (wrong K or tampered)	Send Auth Failure (MAC failure)	AMF informs AUSF; may retry
SQN Sync Failure	SQN out of acceptable range	Send Auth Failure (Synch failure) + AUTS	UDM resynchronizes SQN, generates new AV
Network Auth Failure	UE cannot verify network identity	Abort authentication	—
RES* Mismatch	UE response doesn't match expected	—	AUSF reports failure; AMF rejects registration
HXRES* Mismatch	Initial check at AMF fails	—	AMF still forwards to AUSF for final decision
AUSF Timeout	AUSF unreachable	—	AMF may select alternate AUSF or reject
SUCI Decryption Failure	Invalid SUCI format or wrong HN key	—	UDM reports error; registration fails

3 Security Establishment

Purpose

The Security Establishment procedure activates NAS and AS security between the UE and the network. It selects and activates ciphering and integrity protection algorithms for both NAS signaling (between UE and AMF) and AS signaling/user plane (between UE and gNB). This procedure ensures all subsequent communications are protected against eavesdropping and tampering.

Preconditions

- Authentication has been successfully completed
- K_{AMF} has been derived at both UE and AMF
- UE Security Capabilities have been provided to the AMF
- ngKSI is established (identifies the active security context)

NFs Involved

Network Function	Role
UE	Derives NAS/AS keys, verifies algorithm selection
AMF	Selects NAS algorithms, sends Security Mode Command, derives NAS keys
gNB	Selects AS algorithms, activates RRC/UP security

Messages

Step	Interface	Message	Key Parameters
1	NAS	Security Mode Command	NAS algorithms, ngKSI, Replayed UE Sec Cap, IMEISV request
2	NAS	Security Mode Complete	IMEISV (if requested), NAS message container
3	N2	Initial Context Setup Request	K_{gNB} , UE Security Capabilities
4	RRC	SecurityModeCommand	AS algorithms (ciphering + integrity)
5	RRC	SecurityModeComplete	—
6	N2	Initial Context Setup Response	—

Step-by-Step Flow

- NAS Algorithm Selection:** AMF selects NAS security algorithms based on UE Security Capabilities and AMF policy (highest priority algorithm supported by both)
- NAS Key Derivation (AMF):** AMF derives:
 - $K_{NASint} = \text{KDF}(K_{AMF}, \text{NAS-int-alg-ID}, \text{alg-type})$
 - $K_{NASenc} = \text{KDF}(K_{AMF}, \text{NAS-enc-alg-ID}, \text{alg-type})$
- Security Mode Command:** AMF sends NAS Security Mode Command (integrity protected with K_{NASint} but NOT ciphered):
 - Selected NAS ciphering algorithm (e.g., NEA1, NEA2)
 - Selected NAS integrity algorithm (e.g., NIA1, NIA2)
 - ngKSI, Replayed UE Security Capabilities, IMEISV request
- UE Verification:** UE verifies:
 - Replayed UE Security Capabilities match what was sent (anti-MITM)
 - Integrity protection of the message using derived K_{NASint}
- NAS Key Derivation (UE):** UE derives K_{NASint} and K_{NASenc} using same KDF

6. **Security Mode Complete:** UE sends NAS Security Mode Complete (integrity protected AND ciphered)
7. **NAS Security Active:** NAS security context is now active in both directions
8. **K_{gNB} Derivation:** AMF derives $K_{gNB} = \text{KDF}(K_{AMF}, \text{NAS UL COUNT})$ and sends to gNB via Initial Context Setup Request
9. **AS Algorithm Selection:** gNB selects AS algorithms based on UE Security Capabilities and gNB policy
10. **AS Key Derivation:** gNB derives:
 - $K_{RRCint} = \text{KDF}(K_{gNB}, \text{RRC-int-alg-ID})$
 - $K_{RRCenc} = \text{KDF}(K_{gNB}, \text{RRC-enc-alg-ID})$
 - $K_{UPenc} = \text{KDF}(K_{gNB}, \text{UP-enc-alg-ID})$
 - $K_{UPint} = \text{KDF}(K_{gNB}, \text{UP-int-alg-ID})$ [optional for DRBs]
11. **RRC Security Mode Command:** gNB sends RRC SecurityModeCommand to UE (integrity protected)
12. **UE AS Key Derivation:** UE derives same AS keys from K_{gNB}
13. **RRC Security Mode Complete:** UE responds with RRC SecurityModeComplete (ciphered + integrity protected)
14. **AS Security Active:** Full AS security is now active

■ Key Hierarchy Diagram

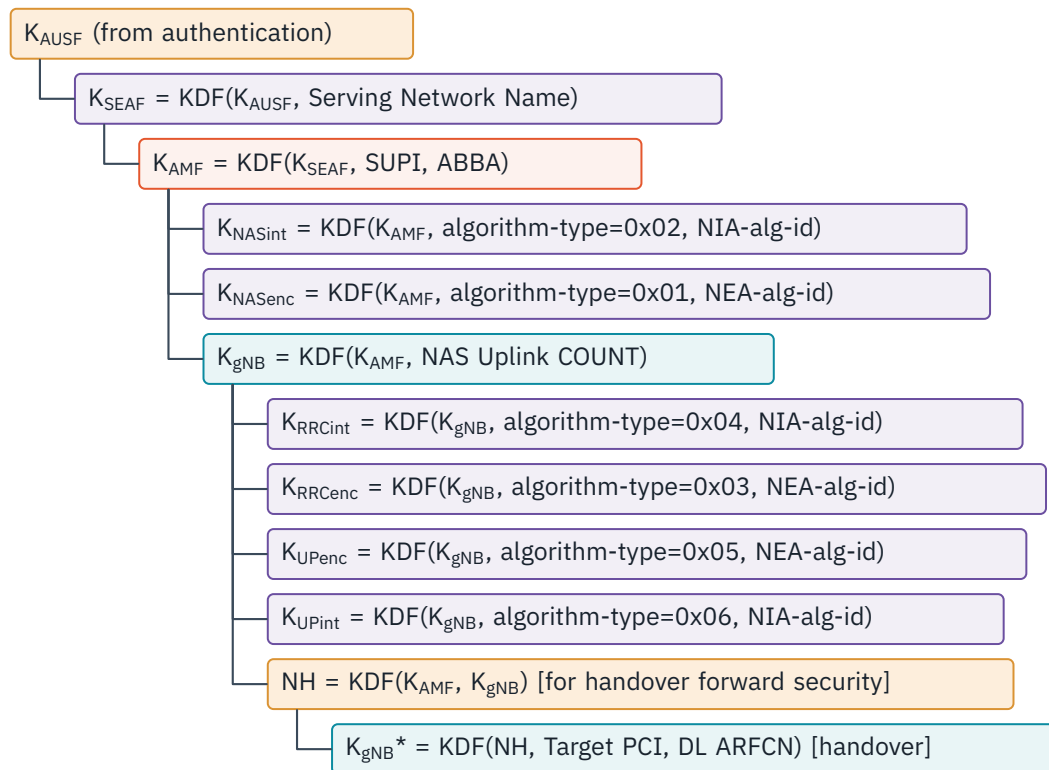


Figure 20.7 Derivation of the NAS and AS keys

Condensed Sequence Diagram

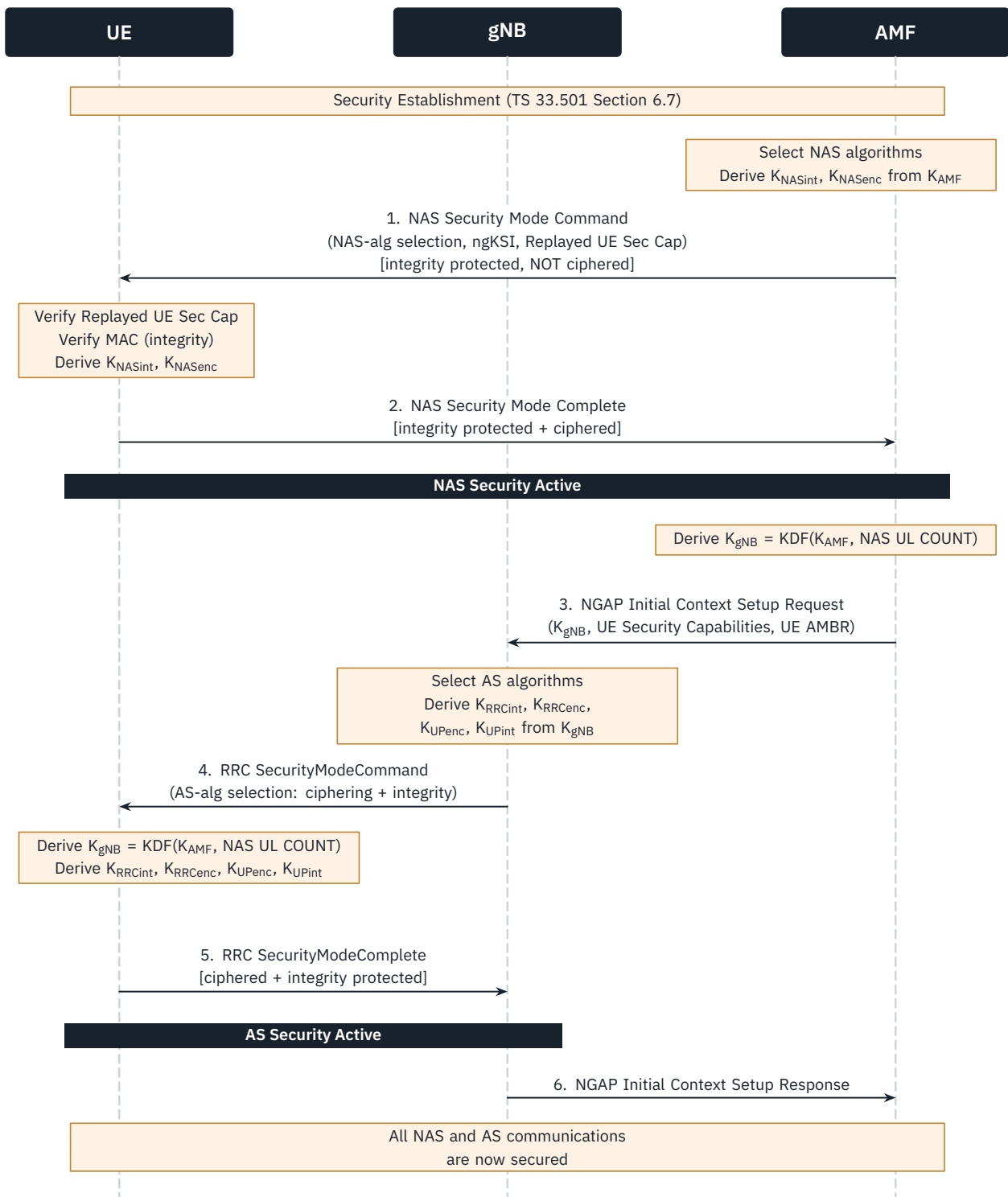


Figure 20.8 NAS and AS security establishment

Failure Cases

Failure	Cause	Response
UE Sec Cap Mismatch	Replayed capabilities don't match (bidding-down attack)	UE sends Security Mode Reject
Integrity Verification Failure	MAC check fails on Security Mode Command	UE sends Security Mode Reject
Algorithm Not Supported	Selected algorithm not in UE capabilities	UE sends Security Mode Reject
NAS Security Mode Reject	Any NAS SMC failure	AMF may re-attempt or reject registration
AS Security Mode Failure	RRC SMC integrity check fails	gNB reports failure; may release connection
Null Integrity Not Allowed	NIA0 selected when not permitted	UE rejects (NIA0 only for emergency)

4 PDU Session Establishment

Purpose

The PDU Session Establishment procedure creates a data connectivity session between the UE and a Data Network (DN). It allocates an IP address (or other PDU type), sets up QoS rules and flows, establishes GTP-U tunnels on N3 (gNB↔UPF) and N9 (UPF↔UPF) interfaces, and configures PFCP sessions on N4 (SMF↔UPF). This enables the UE to send and receive user plane data.

Preconditions

- UE is registered with the 5GC (Registration procedure completed)
- UE is in CM-CONNECTED state (RRC connection active)
- NAS security context is established
- UE has valid subscription for the requested DNN and S-NSSAI
- SMF and UPF resources are available for the requested slice

■ NFs Involved

Network Function	Role
UE	Initiates session request, provides DNN/S-NSSAI/PDU type
gNB	Forwards NAS, establishes N3 GTP-U tunnel
AMF	Routes SM NAS messages, selects SMF
SMF	Session management, IP allocation, QoS policy, UPF selection/control
UPF	User plane forwarding, packet detection/enforcement
PCF	Provides SM policies and PCC rules
UDM	Provides session management subscription data
DN-AAA	External authentication (if required by DN)
CHF	Charging function

■ Messages

Step	Interface	Message
1	NAS	PDU Session Establishment Request (in UL NAS Transport)
2	N11	Nsmf_PDUSession_CreateSMContext Request
3	N10	Nudm_SDM_Get (SM Subscription Data)
4	N7	Npcf_SMPolicyControl_Create
5	N4	PFCP Session Establishment Request
6	N4	PFCP Session Establishment Response
7	N11	Nsmf_PDUSession_CreateSMContext Response
8	N11	Namf_Communication_N1N2MessageTransfer
9	N2	NGAP PDU Session Resource Setup Request
10	RRC	RRCReconfiguration (DRB setup)
11	RRC	RRCReconfigurationComplete
12	N2	NGAP PDU Session Resource Setup Response
13	N11	Nsmf_PDUSession_UpdateSMContext (N2 Info)
14	N4	PFCP Session Modification Request (DL tunnel info)
15	NAS	PDU Session Establishment Accept

■ Step-by-Step Flow

- 1. UE Request:** UE sends PDU Session Establishment Request containing:
 - PDU Session ID, PDU Type (IPv4/IPv6/IPv4v6/Ethernet/Unstructured)
 - Requested DNN, S-NSSAI, SSC Mode
 - SM PDU DN Request Container (optional)

- Requested QoS Rules and QoS Flow Descriptions
- 2. **NAS Transport:** AMF receives UL NAS Transport message with SM NAS container
- 3. **SMF Selection:** AMF selects SMF based on DNN, S-NSSAI, PLMN, and NRF discovery
- 4. **SM Context Creation:** AMF sends Nsmf_PDUSession_CreateSMContext to selected SMF
- 5. **Subscription Data:** SMF retrieves session management subscription data from UDM (Nudm_SDM_Get)
- 6. **SMF Registration:** SMF registers with UDM for the PDU session
- 7. **Policy Association:** SMF creates SM Policy association with PCF (Npcf_SMPolicyControl_Create)
- 8. **PCF Response:** PCF returns PCC rules, QoS policies, and charging rules
- 9. **UPF Selection:** SMF selects appropriate UPF(s) based on UE location, DNN, slice, and data path requirements
- 10. **IP Allocation:** SMF allocates IP address/prefix (from local pool, UPF pool, or external DHCP/AAA)
- 11. **N4 Session Establishment:** SMF sends PFCP Session Establishment Request to UPF:
 - PDRs (Packet Detection Rules), FARs (Forwarding Action Rules)
 - QERs (QoS Enforcement Rules), URRs (Usage Reporting Rules)
 - UL tunnel endpoint (F-TEID on UPF for N3)
- 12. **UPF Response:** UPF responds with allocated F-TEID (N3 UL endpoint)
- 13. **N1N2 Message Transfer:** SMF sends to AMF:
 - N1 SM container: PDU Session Establishment Accept (IP addr, QoS rules, S-NSSAI)
 - N2 SM container: PDU Session Resource Setup Request Transfer (UPF N3 TEID, QoS Profile)
- 14. **RAN Setup:** AMF forwards to gNB via NGAP PDU Session Resource Setup Request
- 15. **DRB Establishment:** gNB maps QoS flows to DRBs, sends RRCReconfiguration to UE
- 16. **UE Configuration:** UE applies QoS rules, configures DRBs, responds with RRCReconfigurationComplete
- 17. **gNB Response:** gNB sends NGAP PDU Session Resource Setup Response (gNB N3 F-TEID for DL)
- 18. **Path Update:** AMF forwards gNB's DL tunnel info to SMF (Nsmf_PDUSession_UpdateSMContext)
- 19. **N4 Modification:** SMF sends PFCP Session Modification to UPF with DL tunnel endpoint (gNB's F-TEID)
- 20. **Session Active:** End-to-end user plane path is established: UE↔gNB↔UPF↔DN

Detailed Sequence Diagram

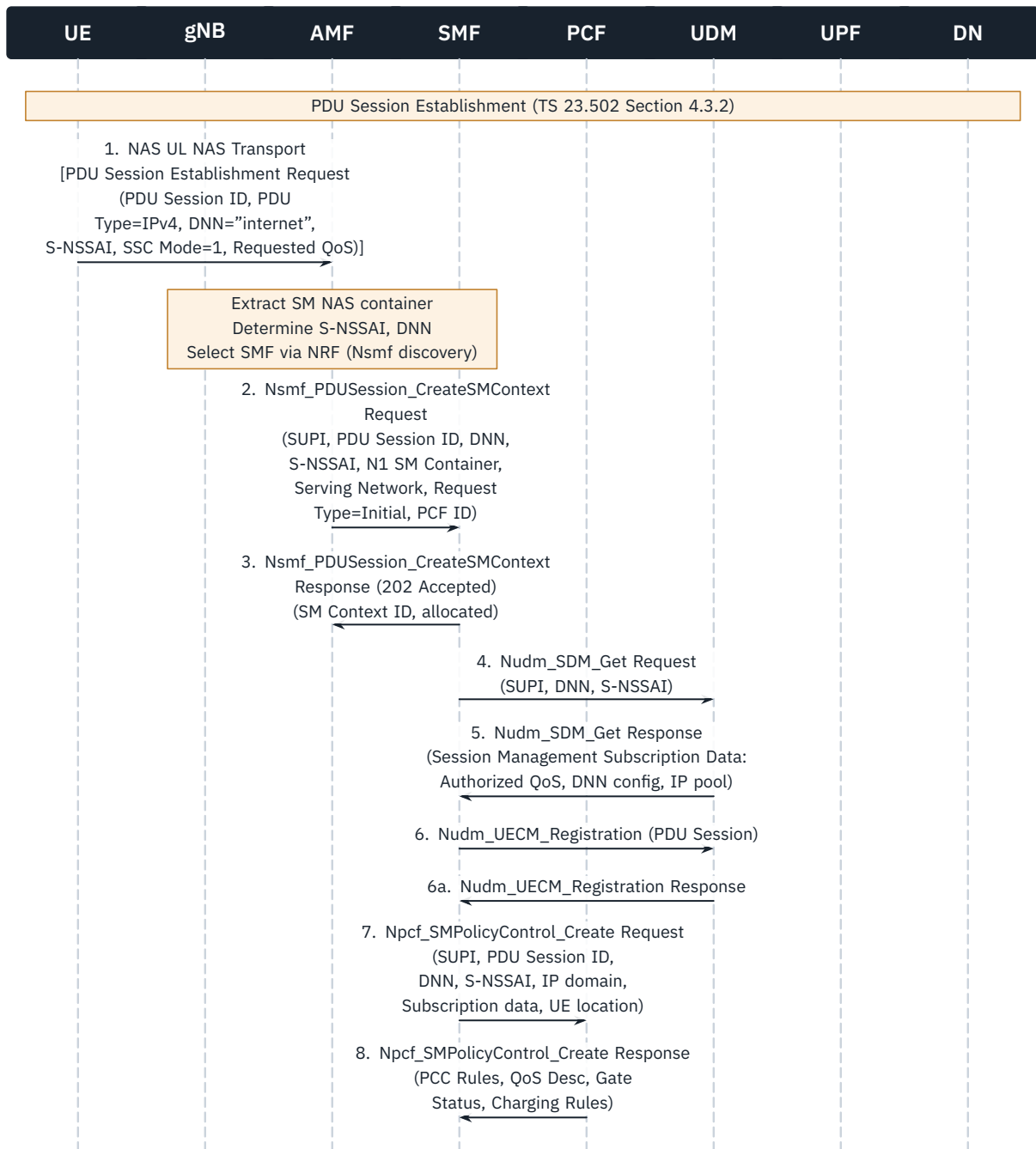


Figure 20.9 PDU session establishment (TS 23.502 §4.3.2) (part 1 of 3)

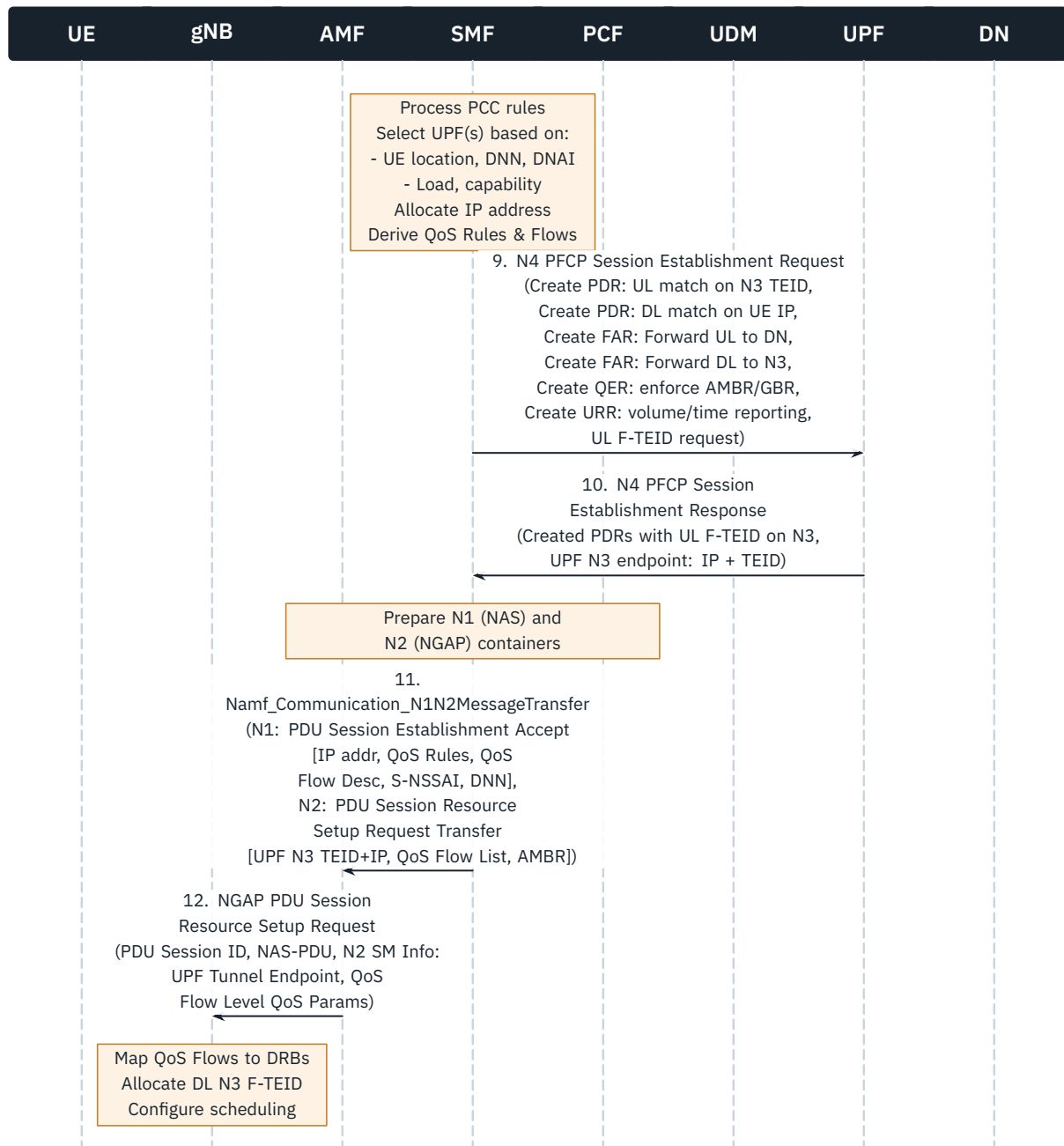


Figure 20.10 PDU session establishment (TS 23.502 §4.3.2) (part 2 of 3)

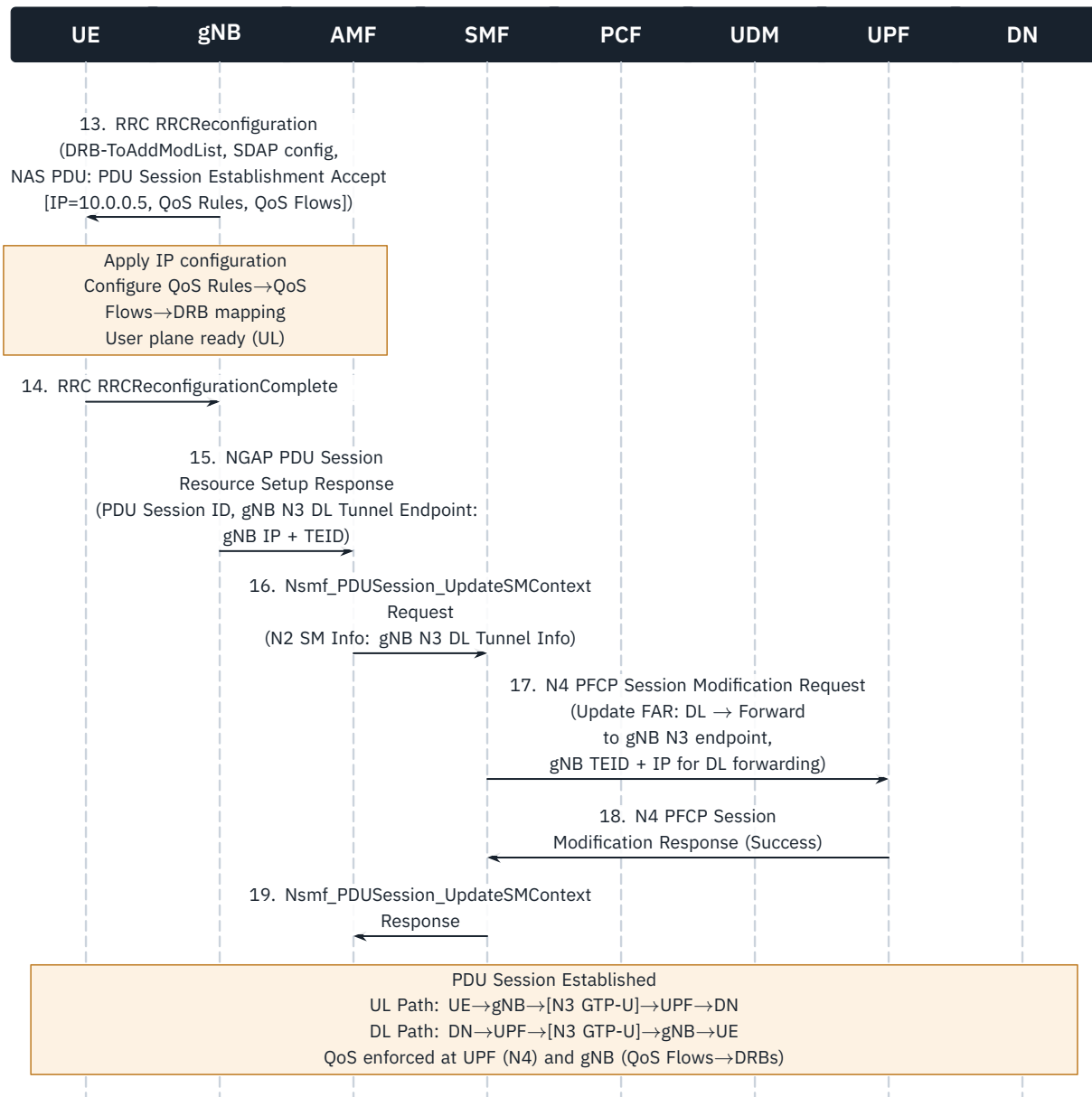


Figure 20.11 PDU session establishment (TS 23.502 §4.3.2) (part 3 of 3)

Failure Cases

Failure	Cause	Response
Insufficient Resources	No IP addresses available or UPF overloaded	PDU Session Establishment Reject (Cause: #26)
DNN Not Subscribed	UE not authorized for requested DNN	Reject (Cause: #27)
Invalid S-NSSAI	Slice not in Allowed NSSAI	Reject (Cause: #62)
SMF Unavailable	No SMF available for DNN/slice	AMF rejects request
UPF Selection Failure	No suitable UPF found	SMF rejects session
PCF Policy Denial	PCF rejects session based on policy	SMF rejects session
N4 Establishment Failure	PFCEP to UPF fails	SMF retries or rejects
DRB Setup Failure	gNB cannot allocate radio resources	Partial/failed resource setup reported to AMF
Max Sessions Reached	UE subscription limit exceeded	Reject (Cause: #65)
Network Failure	Transport network issue	Session establishment timeout

5 PDU Session Release

Purpose

The PDU Session Release procedure tears down an existing PDU session, releasing all associated resources including IP address, GTP-U tunnels (N3/N9), PFCEP sessions (N4), QoS flows, and DRBs. It can be initiated by the UE (user no longer needs connectivity), the SMF (inactivity timer, policy), or the AMF (deregistration).

Preconditions

- An active PDU Session exists (SM context in SMF, N4 session in UPF)
- UE is registered with the network
- NAS security context is active

■ NFs Involved

Network Function	Role
UE	Initiates release (UE-triggered) or acknowledges
gNB	Releases DRBs and N3 tunnel resources
AMF	Routes SM NAS messages
SMF	Orchestrates release, removes SM context
UPF	Removes PFCP session, releases tunnels
PCF	Terminates SM policy association
UDM	Deregisters PDU session
CHF	Final charging record

■ Messages

Step	Interface	Message
1	NAS	PDU Session Release Request (UE-initiated)
2	N11	Nsmf_PDUSession_UpdateSMContext (Release)
3	N4	PFCP Session Deletion Request
4	N4	PFCP Session Deletion Response
5	N7	Npcf_SMPolicyControl_Delete
6	N2	NGAP PDU Session Resource Release Command
7	NAS	PDU Session Release Command
8	RRC	RRCReconfiguration (DRB release)
9	N2	NGAP PDU Session Resource Release Response
10	NAS	PDU Session Release Complete

■ Step-by-Step Flow

■ UE-Initiated Release

1. UE sends PDU Session Release Request (PDU Session ID, Cause)
2. AMF forwards to SMF (Nsmf_PDUSession_UpdateSMContext)
3. SMF initiates resource cleanup
4. SMF sends PFCP Session Deletion Request to UPF
5. UPF releases tunnels, responds with PFCP Session Deletion Response
6. SMF terminates PCF policy association
7. SMF deregisters session from UDM
8. SMF sends PDU Session Release Command to UE (via AMF N1N2 transfer)
9. AMF sends NGAP PDU Session Resource Release Command to gNB
10. gNB releases DRBs, sends RRCReconfiguration to UE
11. gNB responds with PDU Session Resource Release Response

12. UE sends PDU Session Release Complete

■ Network-Initiated Release

1. SMF decides to release (inactivity, policy change, subscription withdrawal)
2. SMF sends PDU Session Release Command to UE via AMF
3. Remaining steps are similar to UE-initiated from step 9

Condensed Sequence Diagram

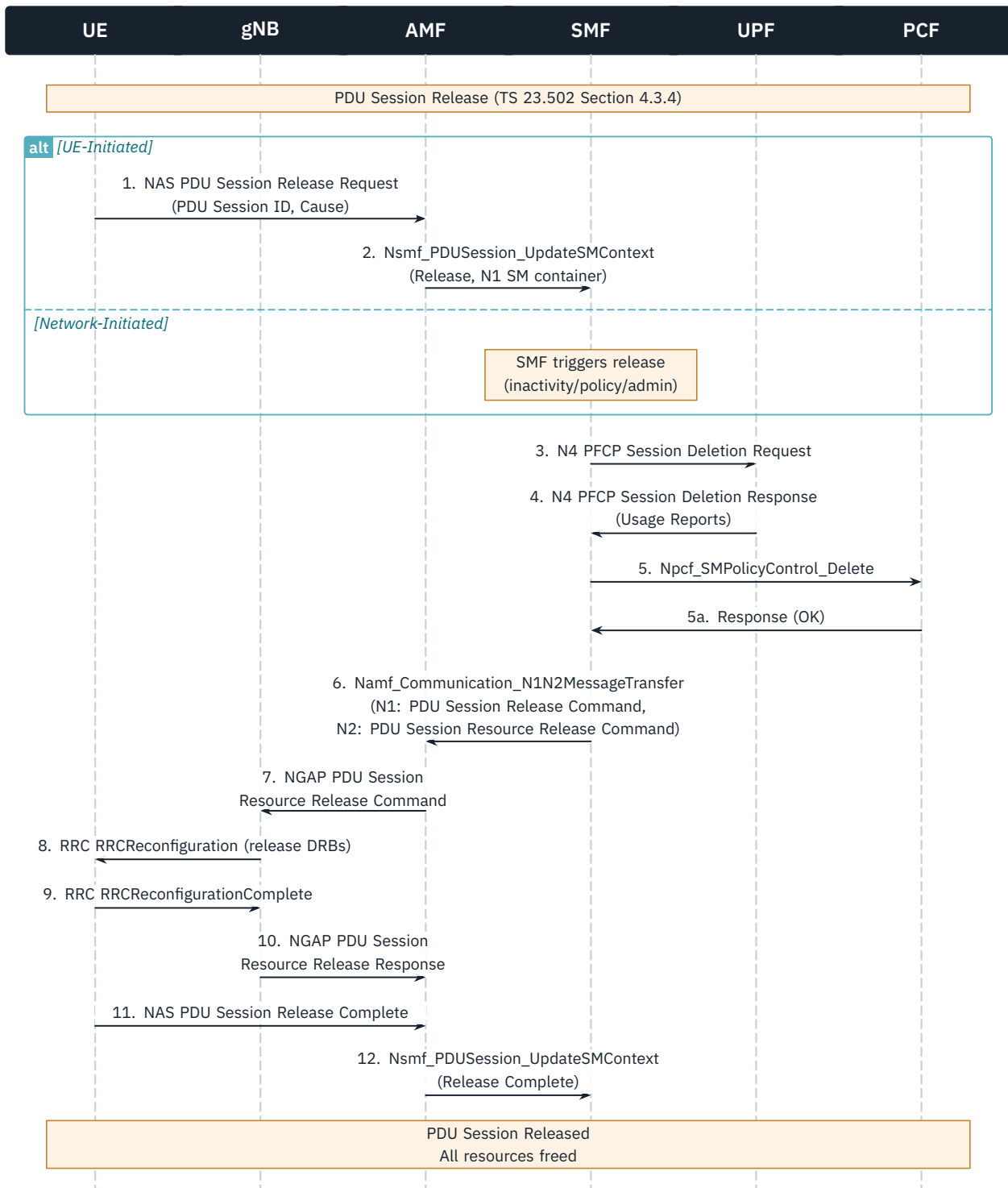


Figure 20.12 PDU session release

Failure Cases

Failure	Cause	Response
UPF Unreachable	N4 path failure	SMF retries; local cleanup after timeout
UE Unreachable (CM-IDLE)	UE in idle mode	SMF proceeds with network-side cleanup
Timer Expiry	No response from UE	Implicit release; local resource cleanup
Partial Release	Some resources fail to release	SMF performs cleanup of remaining resources

6 Service Request

Purpose

The Service Request procedure allows a UE in CM-IDLE state (or CM-CONNECTED with inactive user plane) to re-establish signaling and/or user plane connections when uplink data or signaling needs to be sent. It triggers transition from RRC_IDLE/RRC_INACTIVE to RRC_CONNECTED and reactivates the user plane path (N3 GTP-U tunnels) for existing PDU sessions.

Preconditions

- UE is registered with valid registration context
- UE is in CM-IDLE (RRC_IDLE or RRC_INACTIVE) state
- UE has at least one active PDU session (UP resources to reactivate)
- NAS security context exists (K_{AMF} , K_{NASint} , K_{NASenc} available)
- UE has uplink data or NAS signaling to send

NFs Involved

Network Function	Role
UE	Triggers service request when UL data arrives
gNB	Establishes RRC connection, allocates DRBs
AMF	Processes service request, coordinates UP reactivation
SMF	Updates N4 session with new DL tunnel info
UPF	Receives updated FAR for DL forwarding

Messages

Step	Interface	Message
1	Uu	RRC Resume/Setup Request (if RRC_INACTIVE/IDLE)
2	NAS	Service Request (ngKSI, PDU Session Status)
3	N2	NGAP Initial UE Message or UE Context Resume
4	N2	NGAP Initial Context Setup / PDU Session Resource Setup
5	N11	Nsmf_PDUSession_UpdateSMContext (UP activate)
6	N4	PCF Session Modification (new DL TEID)
7	RRC	RRCReconfiguration (DRB reactivation)
8	NAS	Service Accept

Step-by-Step Flow

- Trigger:** UE application generates uplink data; NAS layer detects CM-IDLE state
- RRC Connection:** UE initiates RRC connection (RRC Setup from IDLE or RRC Resume from INACTIVE)
- Service Request:** UE sends NAS Service Request (integrity protected with K_{NASint}):
 - ngKSI, Service Type (signaling/data/mobile-terminated)
 - PDU Session Status, Uplink Data Status
- AMF Processing:** AMF verifies integrity, identifies active PDU sessions
- Security Activation:** AMF sends new K_{gNB} to gNB (Initial Context Setup or UE Context Resume)
- UP Reactivation:** For each PDU session to reactivate:
 - AMF sends Nsmf_PDUSession_UpdateSMContext to SMF (new AN info needed)
 - gNB allocates DRBs, assigns DL N3 TEID
 - SMF updates UPF with new DL tunnel info (PCF Session Modification)
- DRB Setup:** gNB sends RRCReconfiguration to UE with DRB configuration
- Service Accept:** AMF sends Service Accept to UE
- Data Flow:** User plane path is re-established; buffered DL data is forwarded

Condensed Sequence Diagram

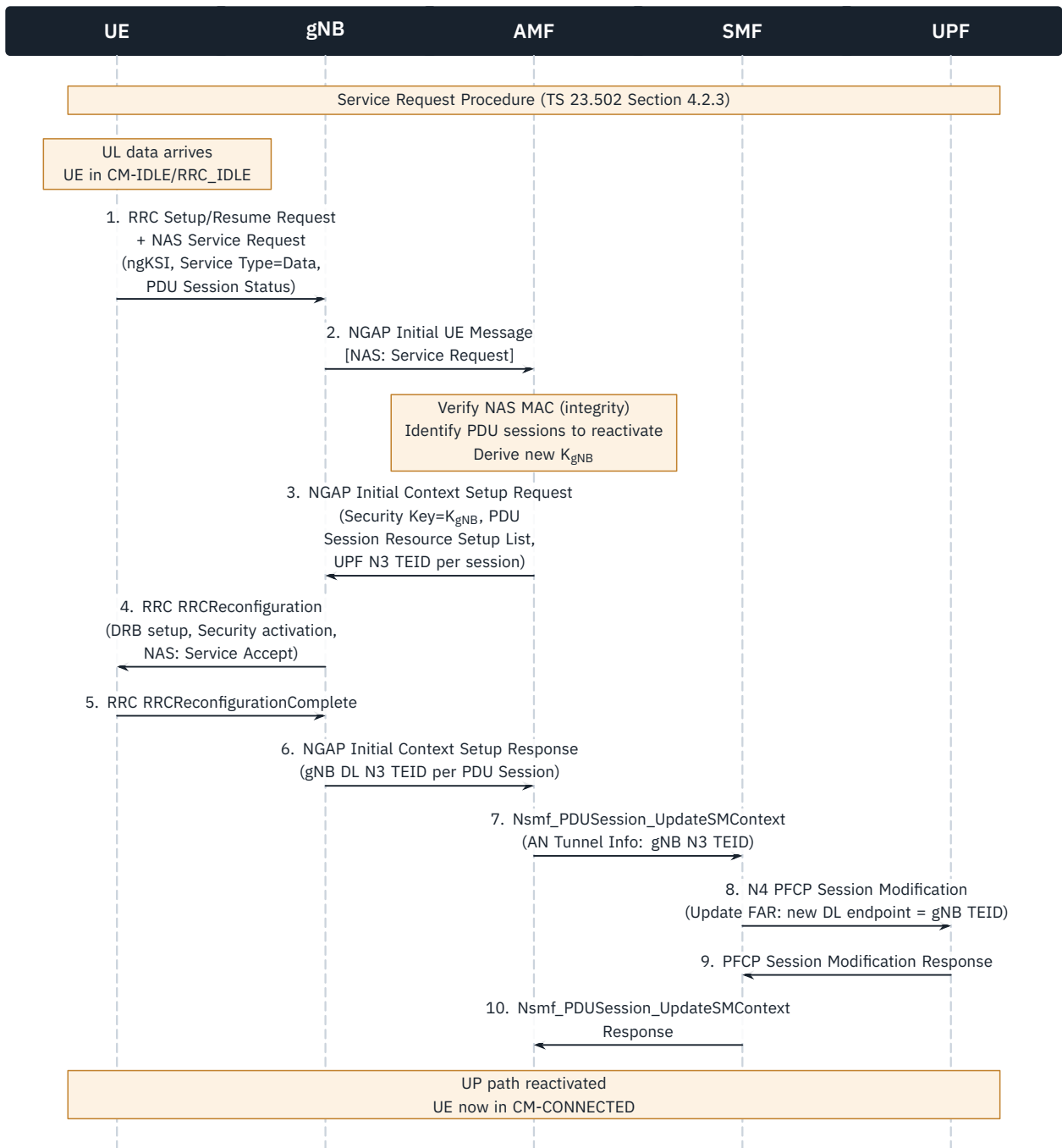


Figure 20.13 Service request

Failure Cases

Failure	Cause	Response
Integrity Failure	NAS MAC verification fails	AMF rejects; may trigger re-authentication
Context Not Found	AMF has no UE context (e.g., AMF restart)	Service Reject (Cause: #9) → UE re-registers
PDU Session Mismatch	Status inconsistency between UE and network	AMF triggers PDU session status sync
RAN Resources Unavailable	gNB cannot allocate DRBs	Partial acceptance or rejection
Security Context Invalid	ngKSI mismatch or expired context	Trigger re-authentication

7 Paging

Purpose

The Paging procedure allows the network to reach a UE that is in CM-IDLE state when downlink data or signaling arrives for that UE. The AMF distributes paging messages to all gNBs in the UE's Registration Area (set of Tracking Areas), and the UE responds with a Service Request to re-establish connectivity.

Preconditions

- UE is in CM-IDLE state (RRC_IDLE or RRC_INACTIVE)
- Downlink data arrives at UPF for the UE, or MT signaling is pending
- AMF has valid UE context with Registration Area information
- UE is monitoring paging occasions in its current cell

■ NFs Involved

Network Function	Role
UPF	Detects DL data for idle UE, sends notification
SMF	Receives data notification, triggers paging via AMF
AMF	Distributes paging to gNBs in Registration Area
gNB(s)	Broadcasts paging message on Uu interface
UE	Monitors paging, initiates Service Request

■ Messages

Step	Interface	Message
1	N4	Downlink Data Notification (UPF→SMF via PFCP)
2	N11	Nsmf_PDUSession_UpdateSMContext (DL data pending) or Namf_Communication_N1N2MessageTransfer
3	N2	NGAP Paging (to all gNBs in Registration Area)
4	Uu	RRC Paging message (5G-S-TMSI)
5	Uu	Service Request (UE response)

■ Step-by-Step Flow

1. **DL Data Arrival:** Downlink packet arrives at UPF for a UE in CM-IDLE state
2. **Data Notification:** UPF buffers the packet and sends Downlink Data Notification to SMF (via N4 PFCP)
3. **Paging Trigger:** SMF sends Namf_Communication_N1N2MessageTransfer or Nsmf notification to AMF indicating DL data pending
4. **Paging Distribution:** AMF sends NGAP Paging message to all gNBs in the UE's Registration Area:
 - 5G-S-TMSI (for UE identification)
 - TAI List (Tracking Area Identities)
 - Paging Priority (if applicable)
 - Paging DRX information
5. **RRC Paging:** Each gNB broadcasts RRC Paging message on PCCH at the UE's paging occasion
6. **UE Detection:** UE wakes up at its paging occasion (determined by 5G-S-TMSI and DRX cycle), detects its identity
7. **Service Request:** UE initiates Service Request procedure (see Section 6)
8. **Data Delivery:** After UP path is reactivated, UPF forwards buffered DL data
9. **Paging Retry:** If UE doesn't respond within T3513, AMF retries paging (up to configured retransmissions)

Condensed Sequence Diagram

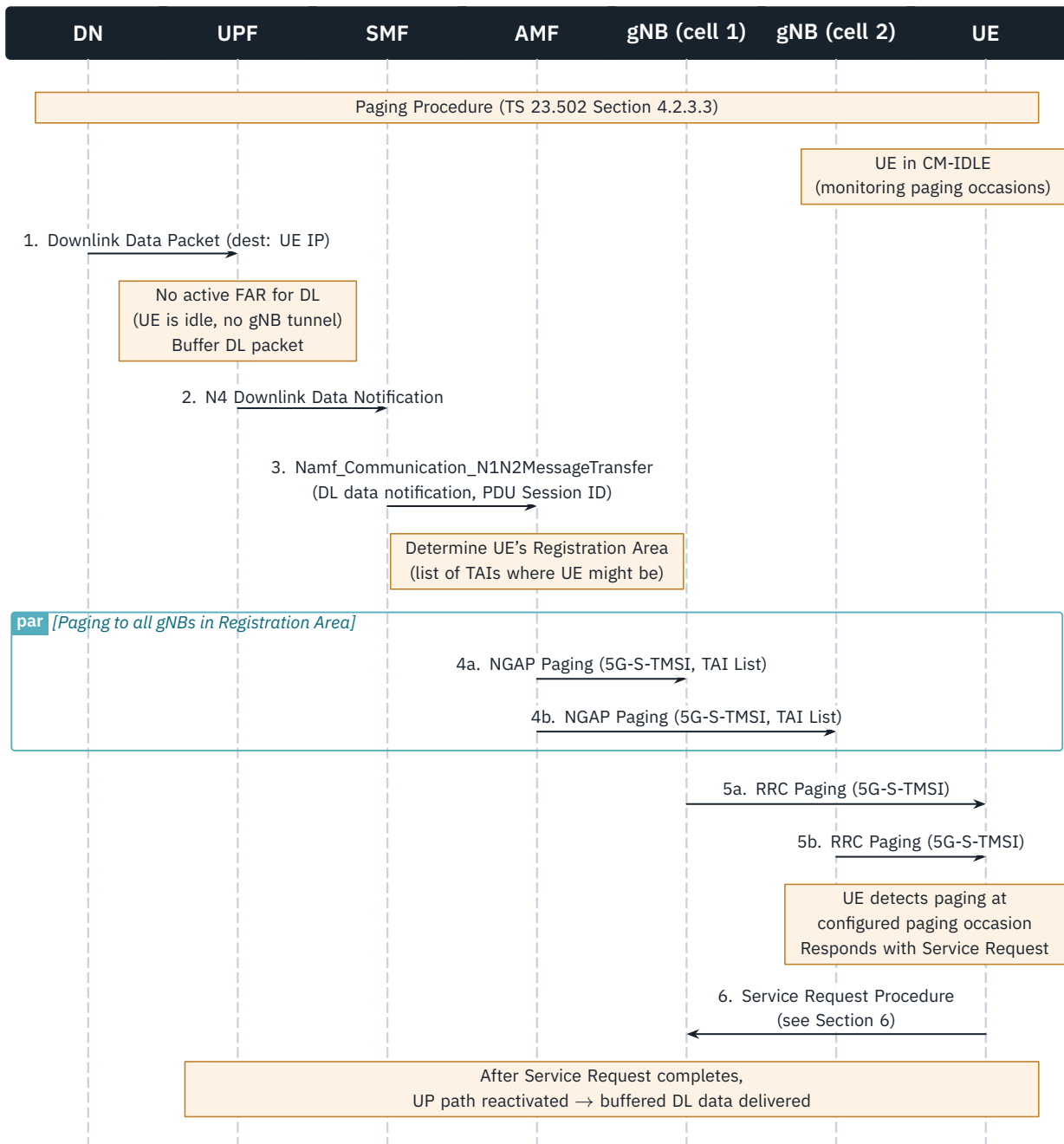


Figure 20.14 Paging of a CM-IDLE UE

Failure Cases

Failure	Cause	Response
UE Not Responding	UE powered off, out of coverage	AMF retries (T3513); eventually notifies SMF
Paging Timeout	Max retransmissions reached	AMF informs SMF; UPF may discard buffered data
Wrong Registration Area	UE moved without updating TA	Paging fails; UE will eventually re-register
UPF Buffer Overflow	Too much buffered DL data	UPF discards oldest packets
Paging Congestion	Too many paging requests	AMF applies paging restrictions/priorities

8 Xn Handover

Purpose

The Xn Handover procedure transfers a UE's connection from a source gNB to a target gNB over the Xn interface without involving the AMF in the handover decision. It provides seamless mobility with minimal service interruption. The source gNB makes the handover decision based on measurement reports, prepares the target gNB, and the user plane path is switched at the UPF. This is used when both gNBs are connected to the same AMF and an Xn interface exists between them.

Preconditions

- UE is in RRC_CONNECTED state with active PDU sessions
- Xn interface exists between source and target gNBs
- Both gNBs are served by the same AMF
- UE measurement reports indicate target cell is better than source
- Target cell belongs to same or compatible network slice
- Source gNB has the UE's security context and active AS keys

NFs Involved

Network Function	Role
UE	Sends measurement reports, performs handover to target
Source gNB	Makes HO decision, prepares target, forwards data
Target gNB	Admits UE, allocates resources, becomes new serving
AMF	Updates UE context with new serving gNB (path switch)
SMF	Updates N4 session for new downlink tunnel
UPF	Switches downlink path to target gNB

Messages

Step	Interface	Message
1	Uu	RRC MeasurementReport (UE→Source gNB)
2	Xn	Handover Request (Source→Target)
3	Xn	Handover Request Acknowledge (Target→Source)
4	Uu	RRC RRCReconfiguration (HO Command)
5	—	UE detaches from source, syncs to target
6	Uu	RRC RRCReconfigurationComplete (UE→Target)
7	Xn	SN Status Transfer (Source→Target)
8	N2	NGAP Path Switch Request (Target→AMF)
9	N11	Nsmf_PDUSession_UpdateSMContext (AMF→SMF)
10	N4	PCF Session Modification (SMF→UPF)
11	N11	Nsmf_PDUSession_UpdateSMContext Response
12	N2	NGAP Path Switch Request Acknowledge (AMF→Target)
13	Xn	UE Context Release (Target→Source)

Step-by-Step Flow

- Measurement Configuration:** Source gNB configures UE with measurement objects and reporting criteria
- Measurement Report:** UE detects target cell signal meets threshold; sends MeasurementReport to source gNB
- HO Decision:** Source gNB evaluates measurement report, selects target cell, decides to handover
- Handover Preparation:**
 - Source gNB sends Handover Request to target gNB over Xn:
 - UE context (security, capabilities, PDU session info)
 - UE History Information
 - Source-to-Target transparent container (RRC config)

5. **Admission Control:** Target gNB performs admission control, allocates resources (DRBs, radio resources)
6. **Handover Request Ack:** Target gNB responds with:
 - Target-to-Source transparent container (RRC HO Command)
 - Allocated DL forwarding tunnels (for data forwarding during HO)
7. **HO Command:** Source gNB sends RRCReconfiguration (mobilityControlInfo) to UE containing:
 - Target cell ID, new C-RNTI
 - Target gNB security algorithm configuration
 - Radio resource configuration for target cell
8. **Data Forwarding:** Source gNB begins forwarding buffered/incoming DL data to target gNB (via Xn forwarding tunnel)
9. **SN Status Transfer:** Source gNB sends PDCP SN Status Transfer to target gNB (for lossless handover on AM DRBs)
10. **Random Access:** UE synchronizes to target cell (RACH procedure)
11. **HO Complete:** UE sends RRCReconfigurationComplete to target gNB
12. **Path Switch:** Target gNB sends NGAP Path Switch Request to AMF:
 - New security context info (NH/NCC if needed)
 - New N3 tunnel info for each PDU session (target gNB's F-TEID)
13. **UP Path Update:** AMF sends Nsmf_PDUSession_UpdateSMContext to SMF with new AN tunnel info
14. **UPF Update:** SMF sends PFCP Session Modification to UPF → update DL FAR to point to target gNB
15. **End Marker:** UPF sends end marker packet on old N3 tunnel (to source gNB) to indicate DL path switched
16. **Path Switch Ack:** AMF sends NGAP Path Switch Request Acknowledge to target gNB (new NH, NCC for future HO)
17. **UE Context Release:** Target gNB sends UE Context Release to source gNB over Xn
18. **Source Cleanup:** Source gNB releases UE context and resources

Detailed Sequence Diagram

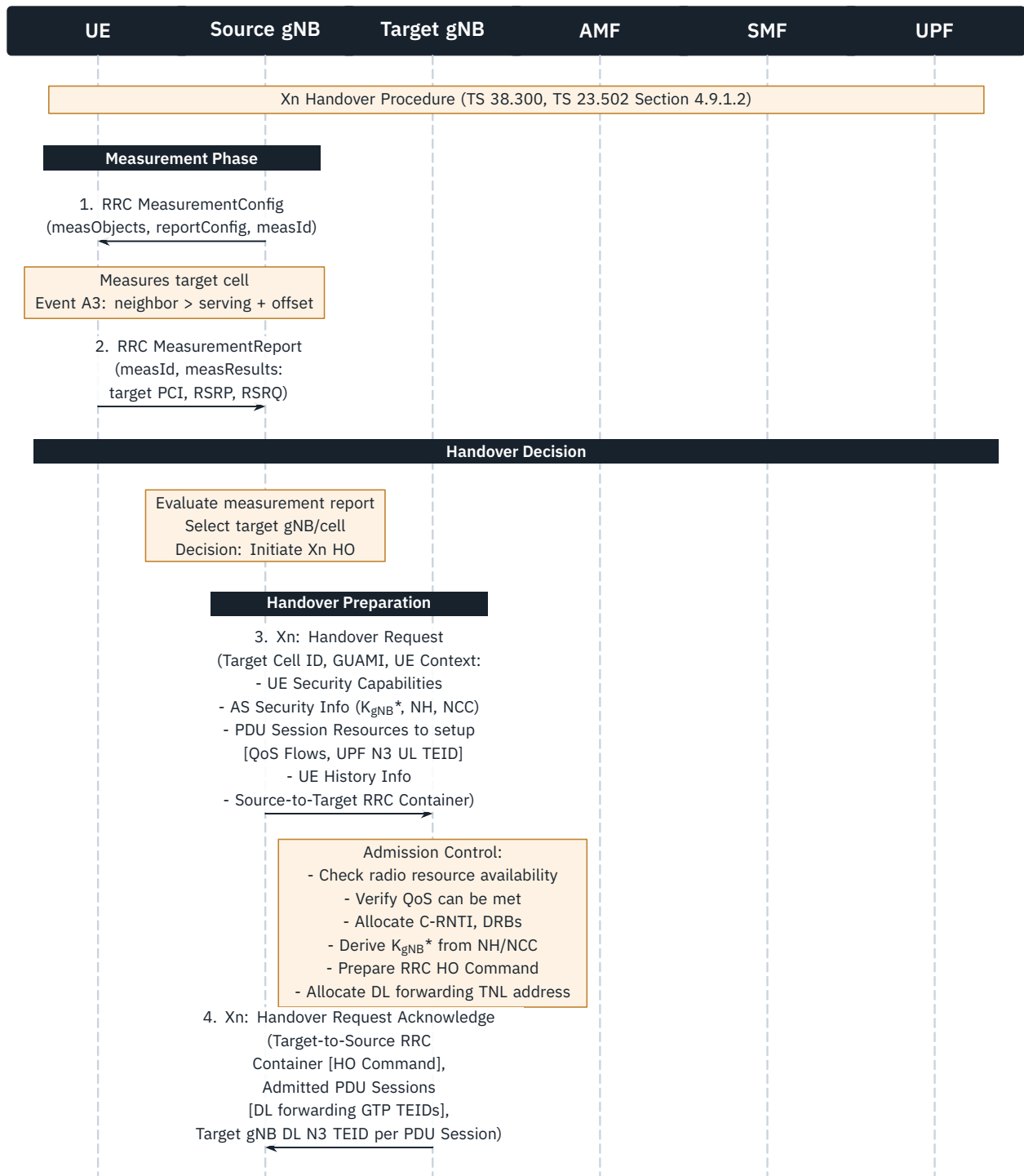


Figure 20.15 Xn handover (part 1 of 3)

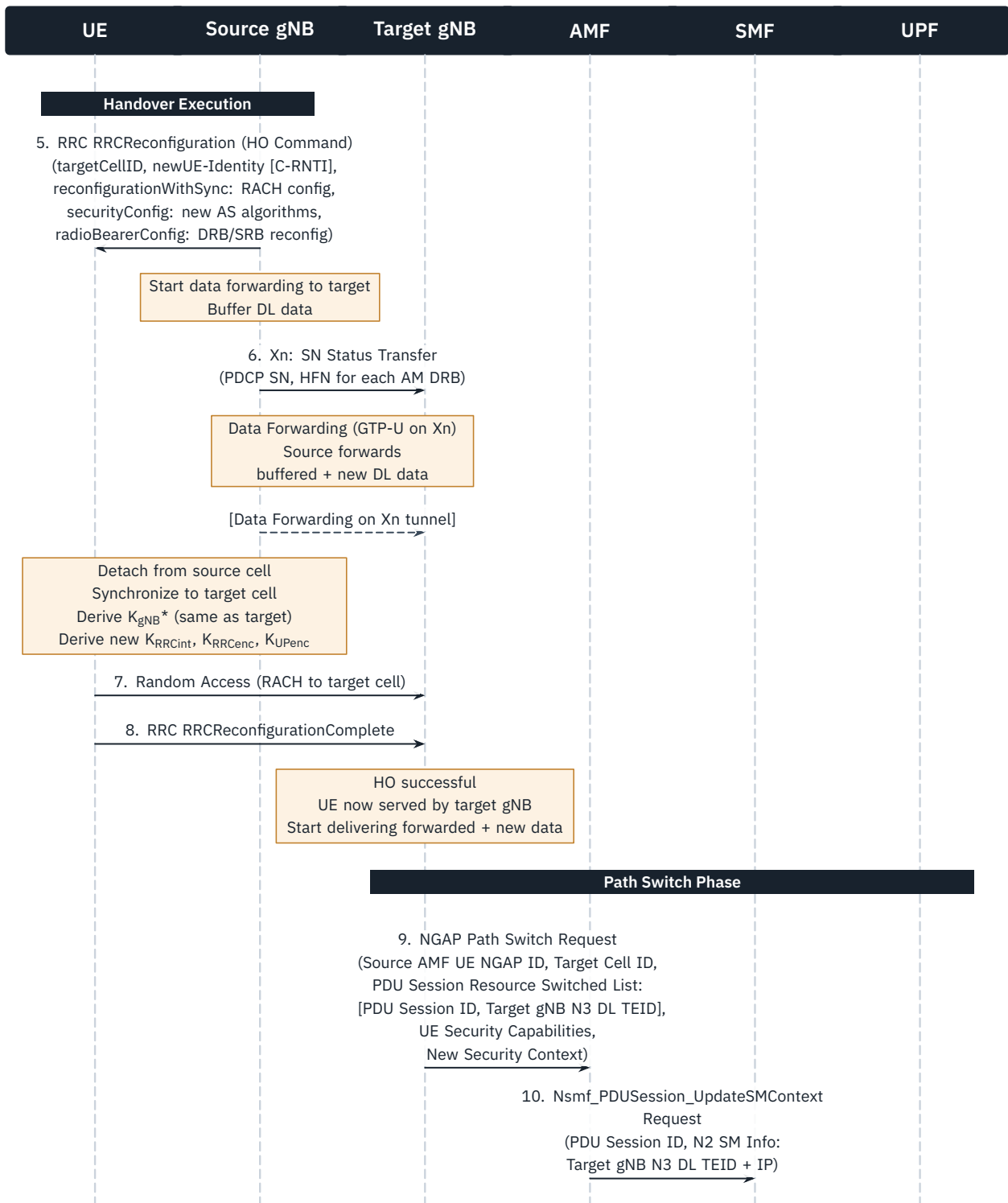


Figure 20.16 Xn handover (part 2 of 3)

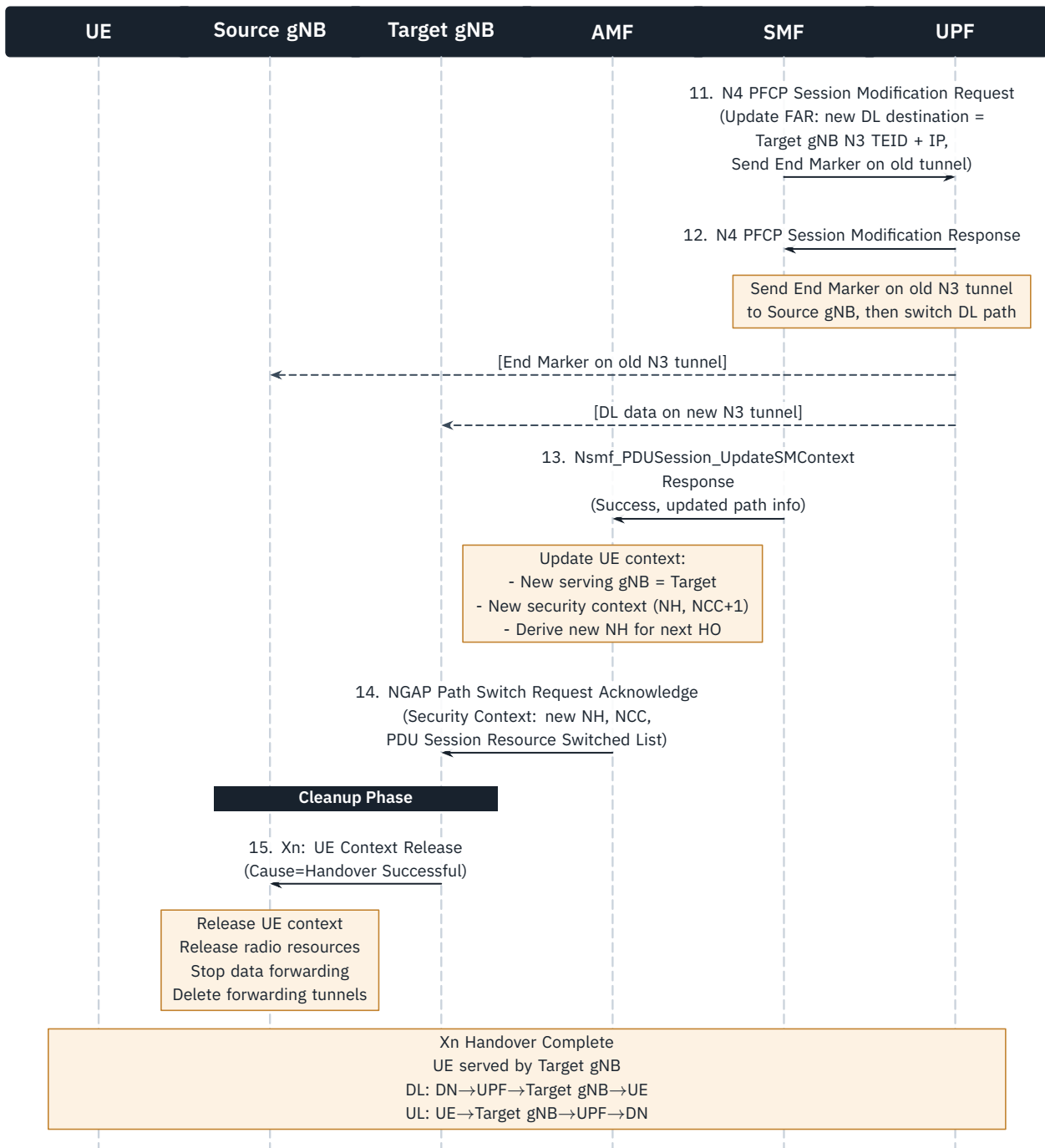


Figure 20.17 Xn handover (part 3 of 3)

Failure Cases

Failure	Cause	Response
HO Preparation Failure	Target rejects (no resources, admission control)	Source gNB tries another target or maintains connection
HO Failure (timer expiry)	UE fails RACH to target within T304	UE initiates RRC Re-establishment to source or other cell
RRC Re-establishment	Radio link failure during HO	UE attempts re-establishment; may trigger new HO
Path Switch Failure	AMF/SMF fails to switch path	AMF sends Path Switch Request Failure; target may release UE
Data Loss	Forwarding tunnel congestion	Some packets lost during HO gap (UM bearers)
Too-Late HO	UE loses source before HO command	RLF; RRC Re-establishment required
Too-Early HO	HO triggered before stable measurement	UE may fail at target; RRC Re-establishment back to source
Xn Interface Failure	Transport issue between gNBs	Fall back to N2 (inter-AMF) handover via AMF

9 N2 Handover (Inter-AMF)

Purpose

The N2 Handover (inter-AMF) procedure handles mobility when the target gNB is served by a different AMF than the source. This is more complex than Xn Handover because it requires UE context transfer between AMFs, re-establishment of NF associations (SMF, PCF), and potentially re-authentication. It is used when no Xn interface exists between source and target gNBs, or when the target falls under a different AMF's serving area.

Preconditions

- UE is in RRC_CONNECTED state with active PDU sessions
- Source gNB determines handover is needed (measurement reports)
- Target gNB is NOT served by the same AMF (different AMF region/set)
- OR no Xn interface exists between source and target gNBs
- N2 interface available between source gNB and source AMF

NFs Involved

Network Function	Role
UE	Performs handover to target cell
Source gNB	Initiates HO, sends HO Required to source AMF
Target gNB	Admits UE, allocates resources
Source AMF	Identifies target AMF, initiates context transfer
Target AMF	Receives UE context, becomes new serving AMF
SMF	Updates UP path for each PDU session
UPF	Switches DL tunnel to target gNB
UDM	Updates AMF registration (new AMF)

Messages

Step	Interface	Message
1	Uu	MeasurementReport (UE→Source gNB)
2	N2	NGAP Handover Required (Source gNB→Source AMF)
3	N14	Namf_Communication_CreateUEContext (Source AMF→Target AMF)
4	N2	NGAP Handover Request (Target AMF→Target gNB)
5	N2	NGAP Handover Request Acknowledge (Target gNB→Target AMF)
6	N14	Namf_Communication_CreateUEContext Response (Target AMF→Source AMF)
7	N2	NGAP Handover Command (Source AMF→Source gNB)
8	Uu	RRC RRCReconfiguration (HO Command to UE)
9	Uu	RRC RRCReconfigurationComplete (UE→Target gNB)
10	N2	NGAP Handover Notify (Target gNB→Target AMF)
11	N14/N2	UE Context Release (cleanup)

■ Step-by-Step Flow

1. **HO Decision:** Source gNB decides handover based on measurement reports
2. **HO Required:** Source gNB sends NGAP Handover Required to source AMF:
 - Target ID (Target gNB + Target TAI)
 - Source-to-Target RRC transparent container
 - PDU Session Resource List
3. **Target AMF Selection:** Source AMF determines target gNB is served by different AMF (based on TAI→AMF mapping)
4. **UE Context Transfer:** Source AMF sends Namf_Communication_CreateUEContext to target AMF:
 - UE context (SUPI, security context, PDU session list)
 - Source-to-Target container
 - MM context, SM contexts
5. **Target gNB Preparation:** Target AMF sends NGAP Handover Request to target gNB
6. **Admission Control:** Target gNB performs admission, allocates resources
7. **HO Request Ack:** Target gNB sends Handover Request Acknowledge (HO Command, DL tunnel info)
8. **Context Response:** Target AMF responds to source AMF with HO Command
9. **HO Command Forwarded:** Source AMF sends NGAP Handover Command to source gNB
10. **RRC HO Command:** Source gNB forwards HO Command to UE via RRCReconfiguration
11. **Data Forwarding:** Source gNB forwards data to target gNB (indirect via UPF or direct if possible)
12. **UE Handover:** UE detaches from source, attaches to target cell
13. **HO Complete:** UE sends RRCReconfigurationComplete to target gNB
14. **HO Notify:** Target gNB sends NGAP Handover Notify to target AMF
15. **Registration Update:** Target AMF registers with UDM as new serving AMF
16. **UP Path Switch:** Target AMF triggers Nsmf_PDUSession_UpdateSMContext for each PDU session
17. **N4 Update:** SMF updates UPF with new DL tunnel info (target gNB TEID)
18. **Source Cleanup:** Target AMF→Source AMF: UE Context Release; Source AMF→Source gNB: UE Context Release Command

Condensed Sequence Diagram

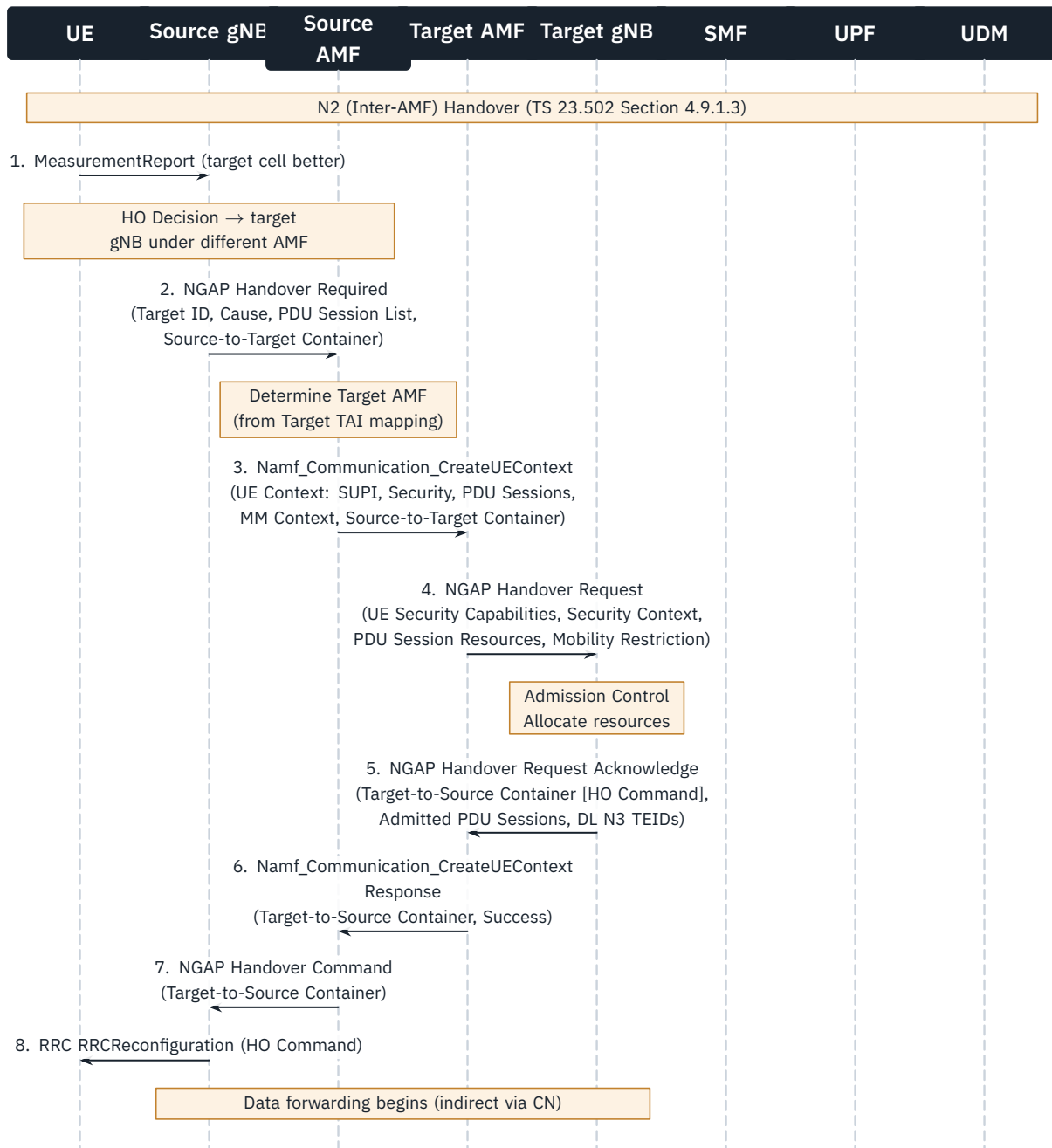


Figure 20.18 N2 (inter-AMF) handover (part 1 of 2)

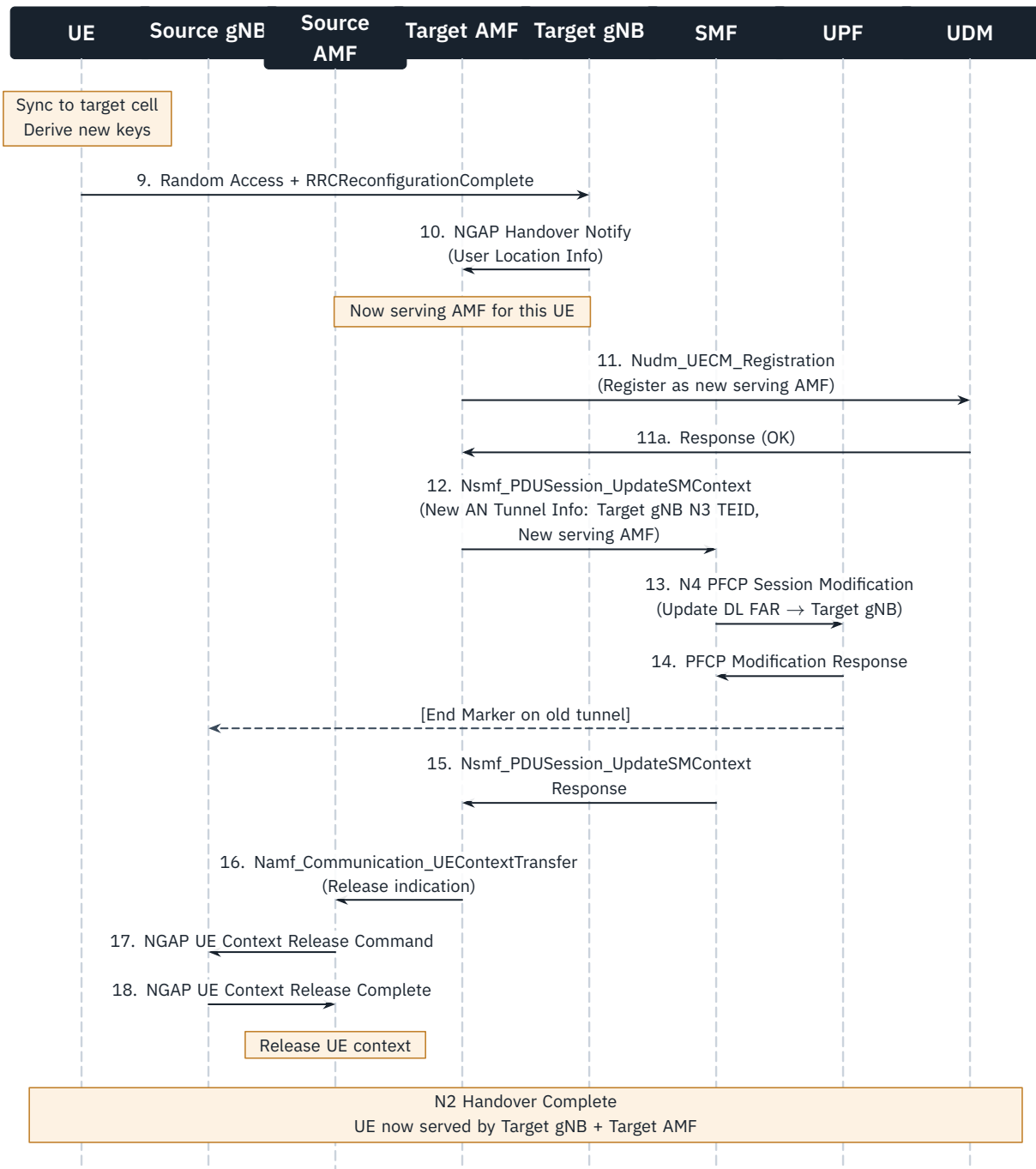


Figure 20.19 N2 (inter-AMF) handover (part 2 of 2)

Failure Cases

Failure	Cause	Response
Target AMF Unreachable	N14 communication failure	Source AMF sends HO Preparation Failure to gNB
Target gNB Rejects	Admission control failure at target	Target AMF informs Source AMF; HO Preparation Failure
HO Failure (UE side)	UE cannot reach target cell	UE attempts RRC Re-establishment
Context Transfer Failure	Incomplete/corrupted UE context	Target AMF rejects; source maintains UE
Timer Expiry (TRELOCprep)	Preparation takes too long	Source AMF cancels HO
Timer Expiry (TRELOCoverall)	Overall HO takes too long	Source AMF releases, target AMF cleans up
Path Switch Failure	SMF/UPF update fails	Handover may succeed but UP path broken; recovery needed
Security Context Mismatch	Keys don't match after transfer	Target AMF triggers re-authentication

Summary: Procedure Comparison Table

Procedure	Trigger	Key NFs	Latency Budget	3GPP Reference
Initial Registration	UE power-on / new PLMN	AMF, AUSF, UDM, PCF	Seconds	TS 23.502 §4.2.2.2
5G-AKA	Registration / Re-auth	AMF, AUSF, UDM	~100ms (NW side)	TS 33.501 §6.1
Security Establishment	Post-authentication	AMF, gNB	~50ms	TS 33.501 §6.7
PDU Session Establishment	UE data request	AMF, SMF, UPF, PCF	~200-500ms	TS 23.502 §4.3.2
PDU Session Release	UE/Network decision	SMF, UPF	~100ms	TS 23.502 §4.3.4
Service Request	UL data in IDLE	AMF, SMF, UPF	~50-100ms	TS 23.502 §4.2.3
Paging	DL data for IDLE UE	AMF, gNBs	~320ms-2s (DRX)	TS 23.502 §4.2.3.3
Xn Handover	Mobility (same AMF)	gNBs, AMF, SMF, UPF	~30-50ms interrupt	TS 38.300 §9.2.3
N2 Handover	Mobility (diff AMF)	gNBs, AMFs, SMF, UPF, UDM	~50-100ms interrupt	TS 23.502 §4.9.1.3

Key References

- **3GPP TS 23.501:** System Architecture for the 5G System (5GS)
- **3GPP TS 23.502:** Procedures for the 5G System (5GS)
- **3GPP TS 33.501:** Security architecture and procedures for 5G System
- **3GPP TS 38.300:** NR; NR and NG-RAN Overall Description
- **3GPP TS 38.331:** NR; Radio Resource Control (RRC) Protocol
- **3GPP TS 29.244:** Interface between the Control Plane and the User Plane (PCF)
- **3GPP TS 38.413:** NG-RAN; NG Application Protocol (NGAP)
- **3GPP TS 38.423:** NG-RAN; Xn Application Protocol (XnAP)